Wazuh SIEM Installation and Nmap Detection

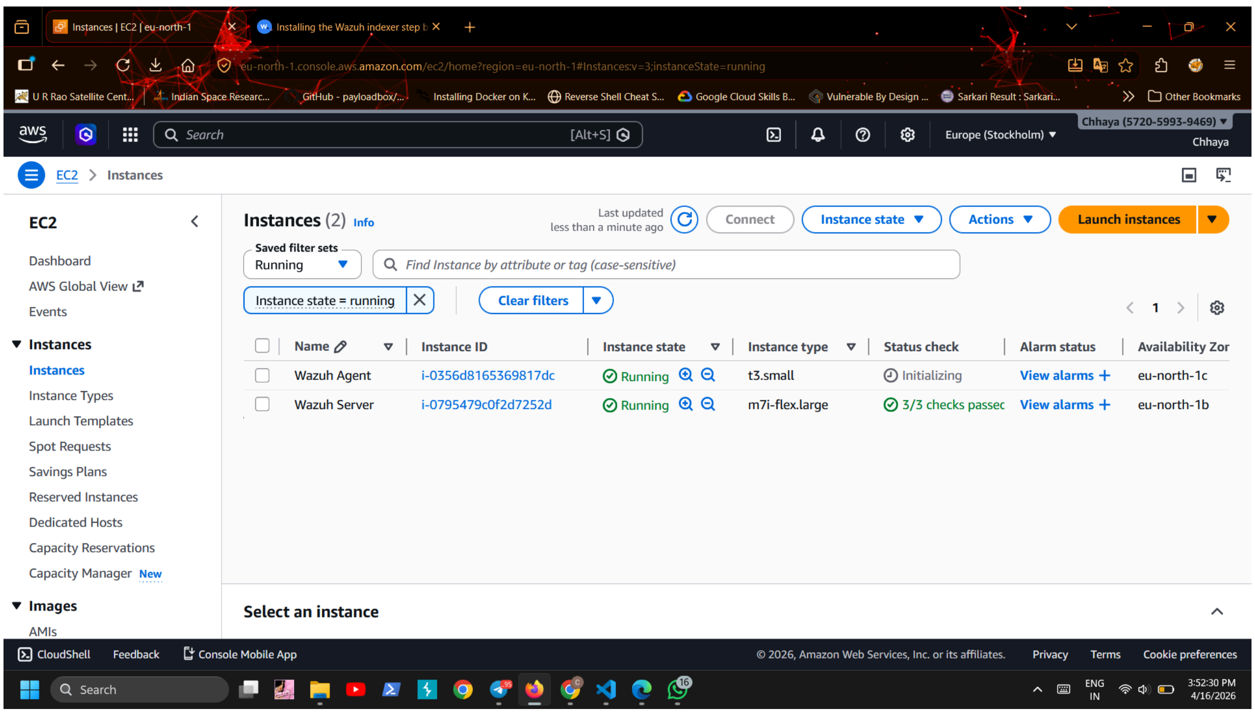
1. Objective

The objective of this task was to install Wazuh SIEM on a local machine (Wazuh Server) and connect another virtual machine (Wazuh Agent) to it. After completing the setup, an Nmap scan was performed on the agent machine to check whether Wazuh detects and generates alerts for such activity.

2. Lab Setup

For this lab, I used two machines:

- Wazuh Server – where Wazuh SIEM is installed
- Wazuh Agent – separate VM connected to the server



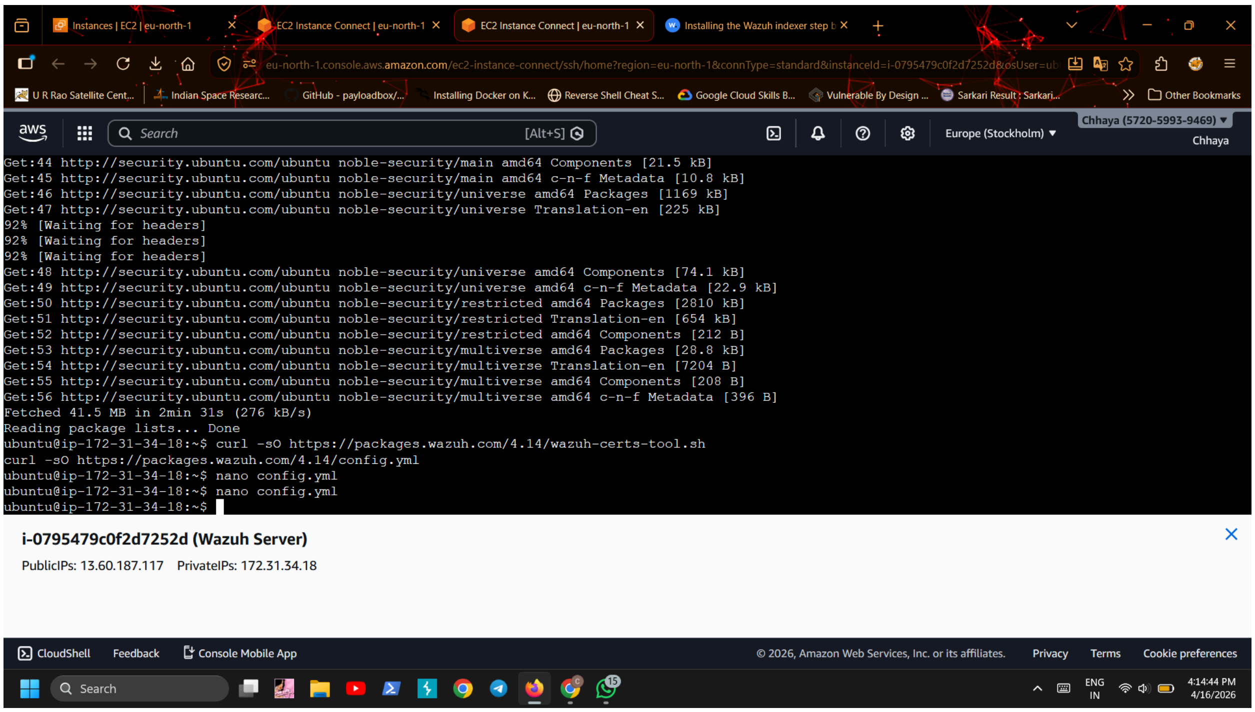
3. Wazuh Installation on Wazuh Server

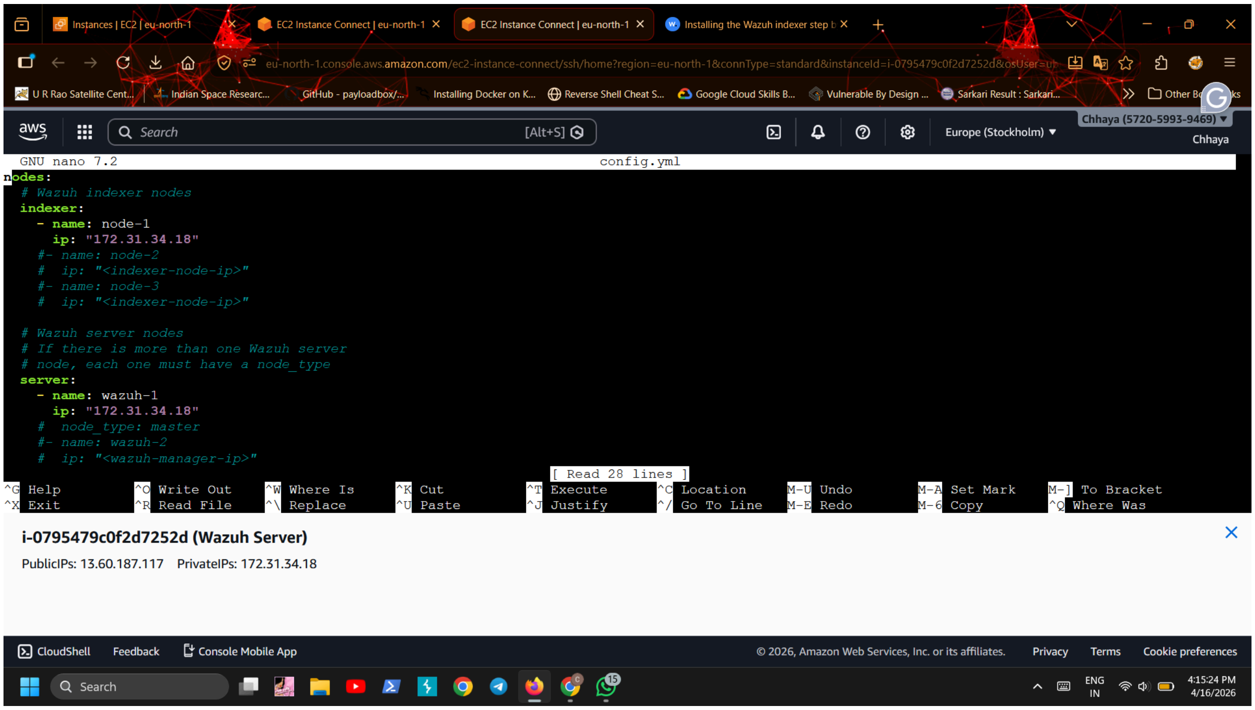
I followed the official Wazuh documentation to install the indexer component. The process was divided into three main stages.

3.1 Certificate Creation

In the first step, I created SSL certificates required for secure communication between Wazuh components.

- Downloaded the certificate generation script and configuration file
- Edited the configuration file to add node names and IP addresses for:
    - Wazuh indexer
    - Wazuh server
    - Wazuh dashboard
- Ran the script to generate certificates
- Compressed the generated certificates into a single file
- Copied the certificate file to the required nodes




GNU nano 7.2
config.yml
nodes:
Wazuh indexer nodes
indexer:
- name: node-1
ip: "172.31.34.18"
#- name: node-2
ip: "<indexer-node-ip>"
#- name: node-3
ip: "<indexer-node-ip>"
Wazuh server nodes
If there is more than one Wazuh server
node, each one must have a node_type
server:
- name: wazuh-1
ip: "172.31.34.18"
node_type: master
#- name: wazuh-2
ip: "<wazuh-manager-ip>"
[Read 28 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was
i-0795479c0f2d7252d (Wazuh Server)
PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18
CloudShell Feedback Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences
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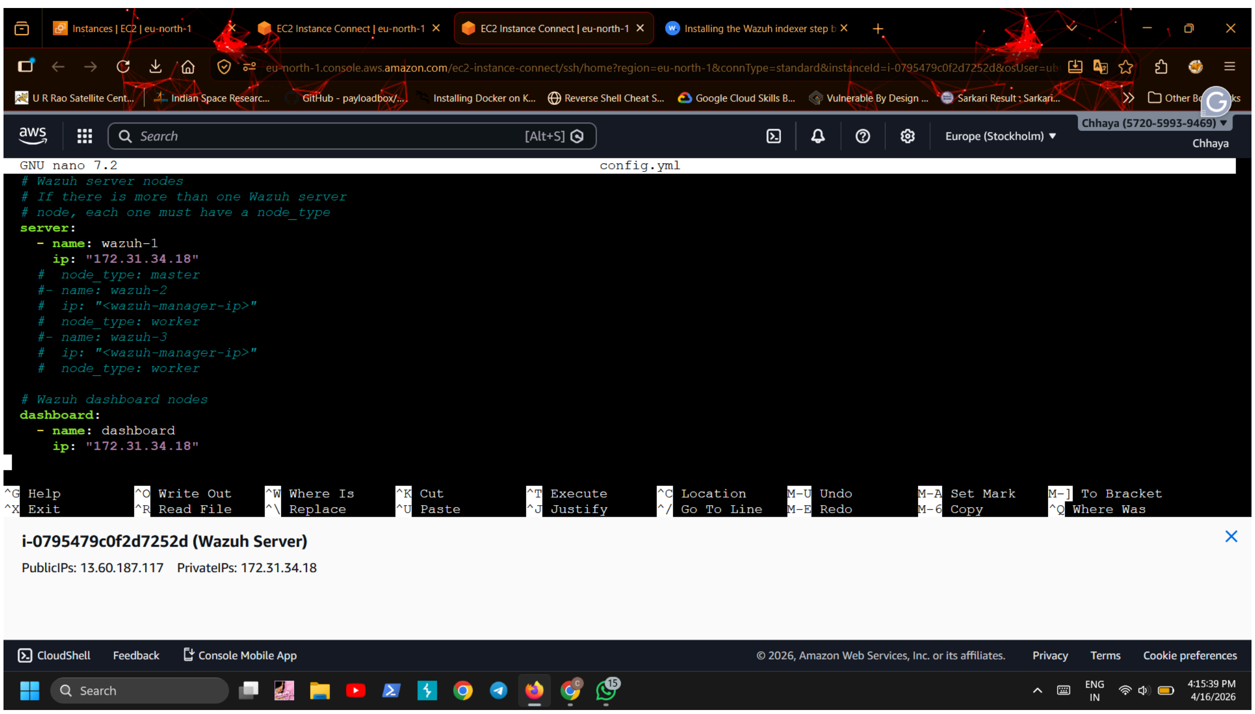
GNU nano 7.2
config.yml
Wazuh server nodes
If there is more than one Wazuh server
node, each one must have a node_type
server:
- name: wazuh-1
ip: "172.31.34.18"
node_type: master
#- name: wazuh-2
ip: "<wazuh-manager-ip>"
node_type: worker
#- name: wazuh-3
ip: "<wazuh-manager-ip>"
node_type: worker
Wazuh dashboard nodes
dashboard:
- name: dashboard
ip: "172.31.34.18"
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was
i-0795479c0f2d7252d (Wazuh Server)
PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18
CloudShell Feedback Console Mobile App
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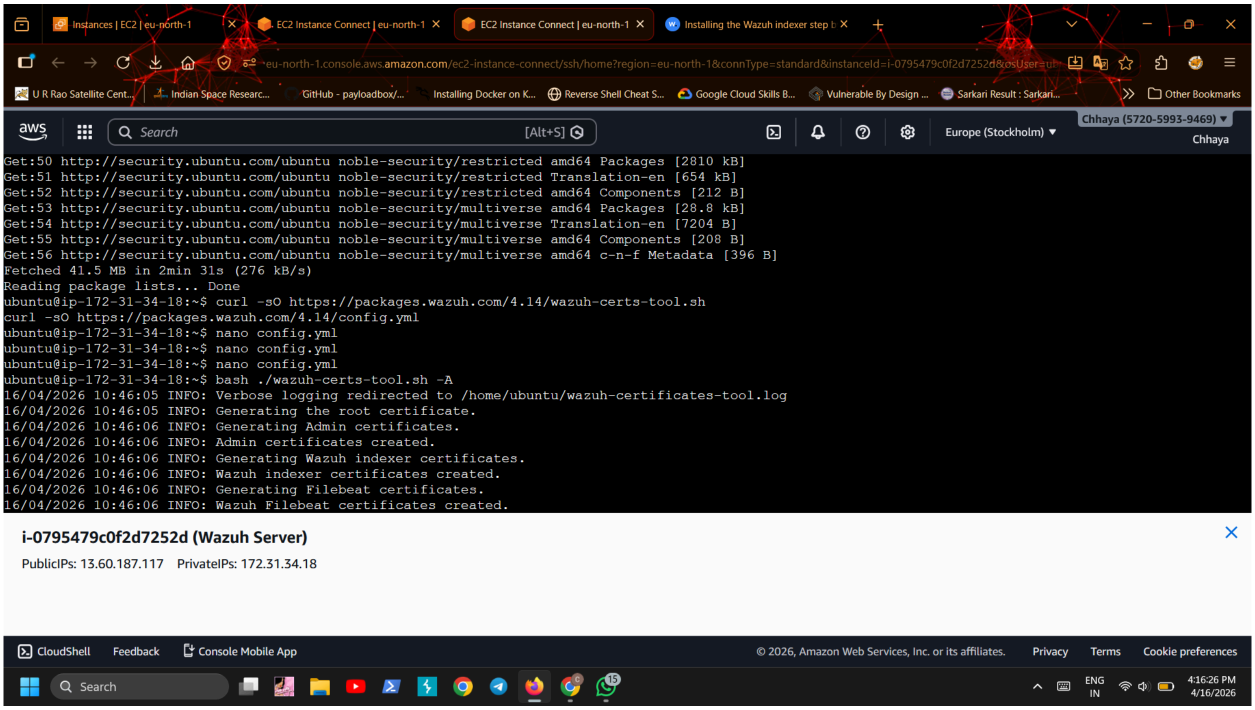
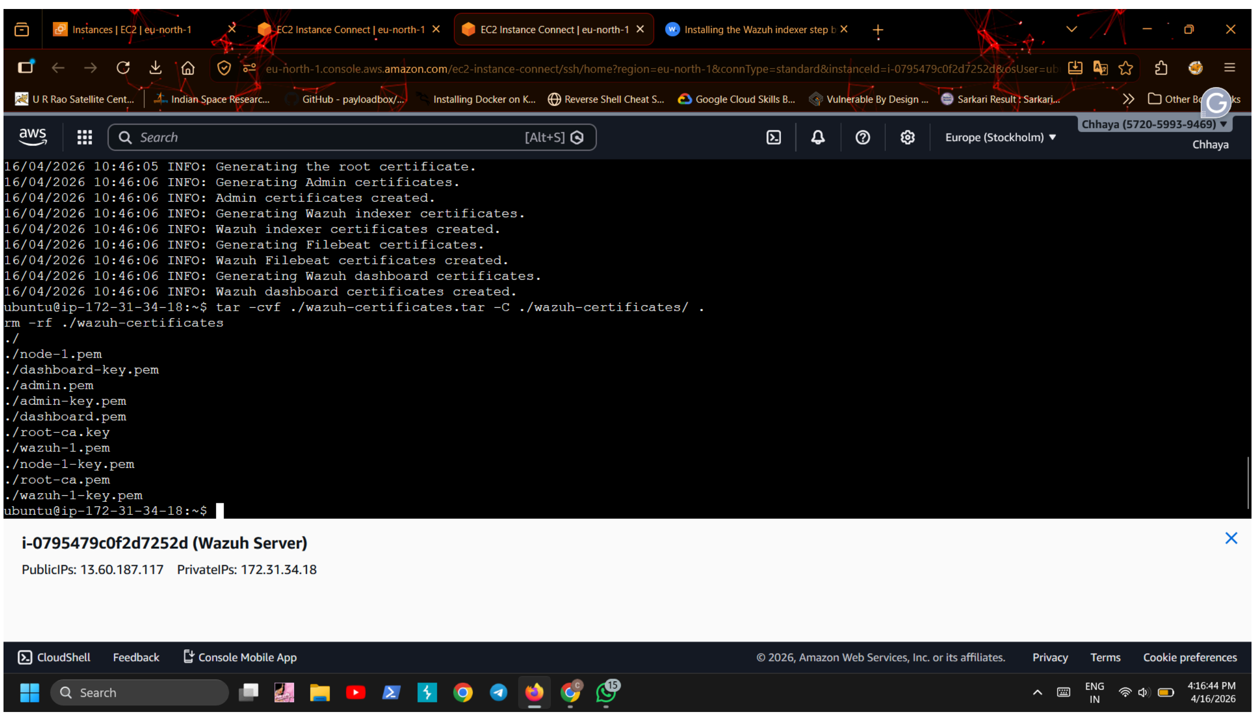
4/16/2026



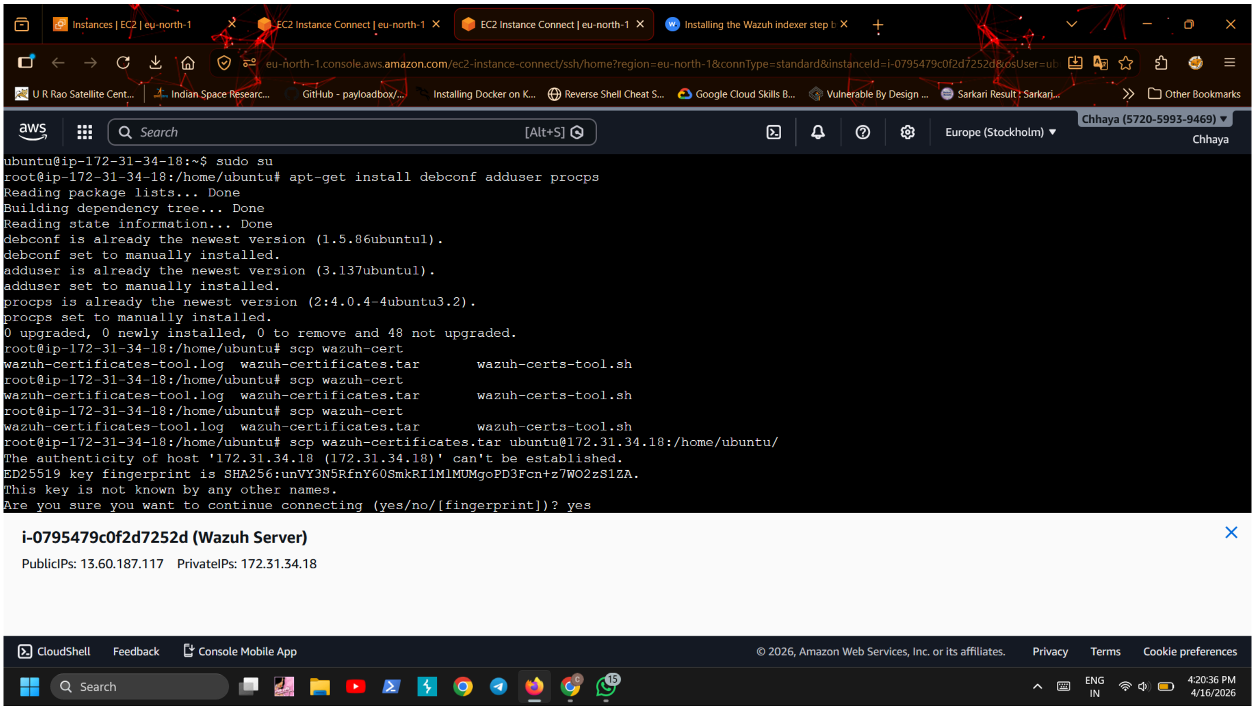


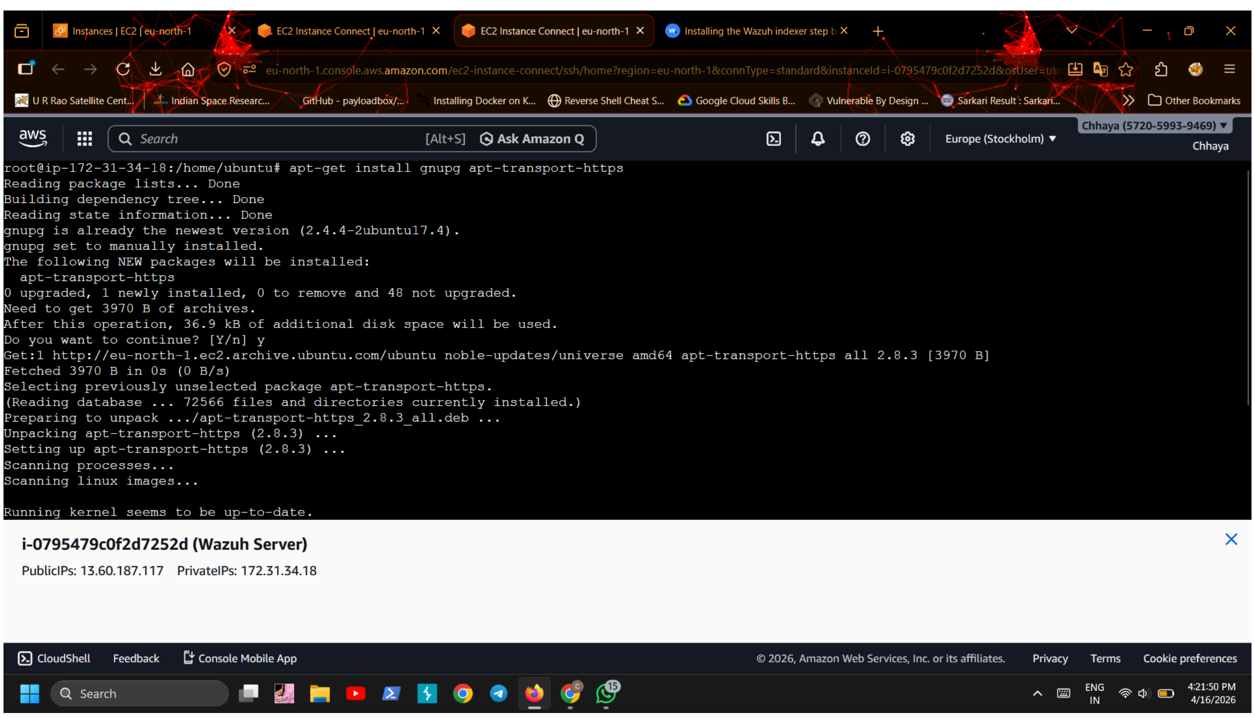
3.2 Wazuh Indexer Installation

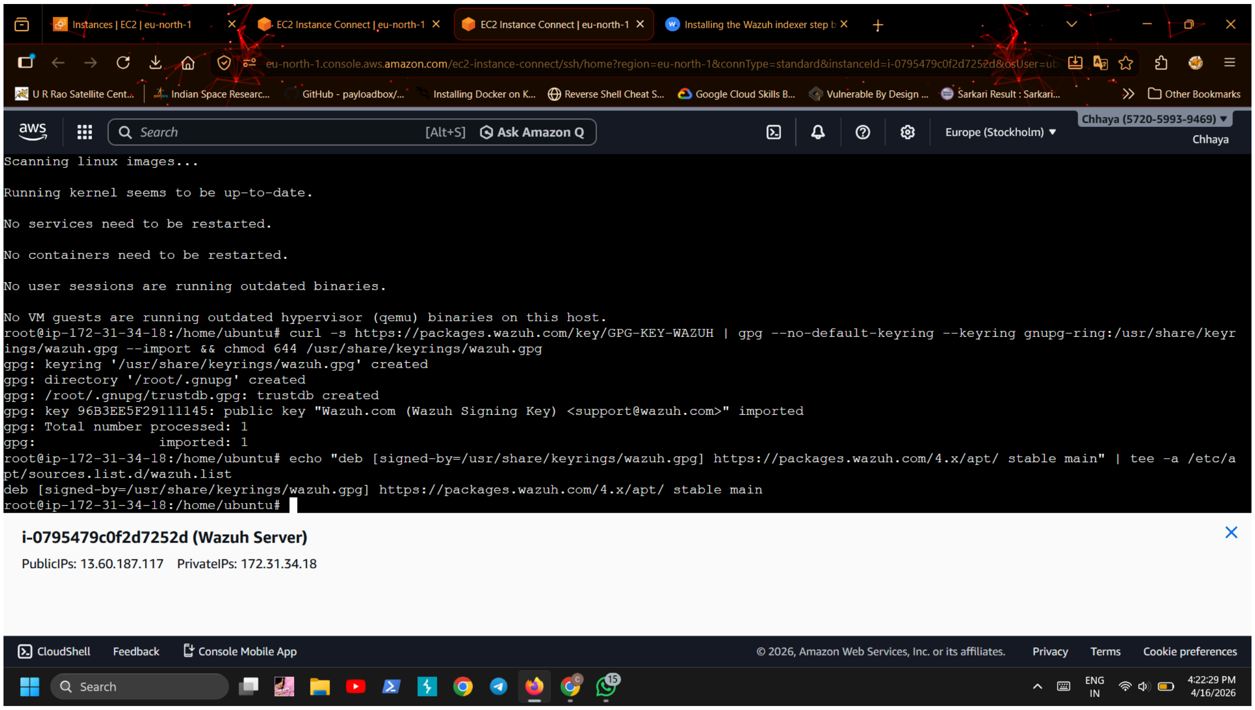
Next, I installed the Wazuh indexer on the Wazuh server.

- Installed required dependencies

- Added the Wazuh repository to the system
- Updated package list
- Installed the Wazuh indexer package






Instances | EC2 | eu-north-1
EC2 Instance Connect | eu-north-1
EC2 Instance Connect | eu-north-1
Installing the Wazuh indexer step b
eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-0795479c0f2d7252d&osUser=ub
Search
[Alt+S]
Ask Amazon Q
Europe (Stockholm)
Chhaya (5720-5993-9469)
Chhaya
Scanning linux images...
Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-34-18:/home/ubuntu# curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg --no-default-keyring --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg --import && chmod 644 /usr/share/keyrings/wazuh.gpg
gpg: keyring '/usr/share/keyrings/wazuh.gpg' created
gpg: directory '/root/.gnupg' created
gpg: /root/.gnupg/trustdb.gpg: trustdb created
gpg: key 96B3EE5F29111145: public key "Wazuh.com (Wazuh Signing Key) <support@wazuh.com>" imported
gpg: Total number processed: 1
gpg: imported: 1
root@ip-172-31-34-18:/home/ubuntu# echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main" | tee -a /etc/apt/sources.list.d/wazuh.list
deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main
root@ip-172-31-34-18:/home/ubuntu#
i-0795479c0f2d7252d (Wazuh Server)
PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18
CloudShell
Feedback
Console Mobile App
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Privacy
Terms
Cookie preferences
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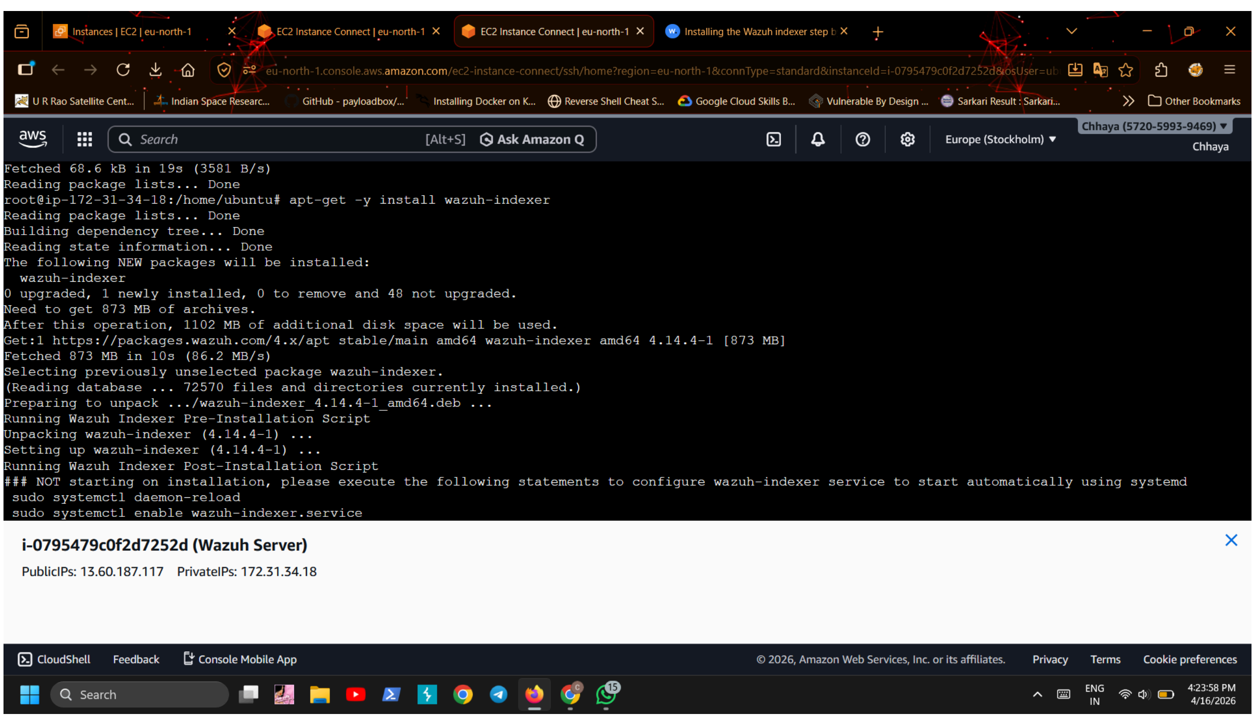
Instances | EC2 | eu-north-1
EC2 Instance Connect | eu-north-1
EC2 Instance Connect | eu-north-1
Installing the Wazuh indexer step b
eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-0795479c0f2d7252d&osUser=ub
Search
[Alt+S]
Ask Amazon Q
Europe (Stockholm)
Chhaya (5720-5993-9469)
Chhaya
Fetched 68.6 kB in 19s (3581 B/s)
Reading package lists... Done
root@ip-172-31-34-18:/home/ubuntu# apt-get -y install wazuh-indexer
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
wazuh-indexer
0 upgraded, 1 newly installed, 0 to remove and 48 not upgraded.
Need to get 873 MB of archives.
After this operation, 1102 MB of additional disk space will be used.
Get:1 https://packages.wazuh.com/4.x/apt stable/main amd64 wazuh-indexer amd64 4.14.4-1 [873 MB]
Fetched 873 MB in 10s (86.2 MB/s)
Selecting previously unselected package wazuh-indexer.
(Reading database ... 72570 files and directories currently installed.)
Preparing to unpack .../wazuh-indexer_4.14.4-1_amd64.deb ...
Running Wazuh Indexer Pre-Installation Script
Unpacking wazuh-indexer (4.14.4-1) ...
Setting up wazuh-indexer (4.14.4-1) ...
Running Wazuh Indexer Post-Installation Script
NOT starting on installation, please execute the following statements to configure wazuh-indexer service to start automatically using systemd
sudo systemctl daemon-reload
sudo systemctl enable wazuh-indexer.service
i-0795479c0f2d7252d (Wazuh Server)
PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18
CloudShell
Feedback
Console Mobile App
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Terms
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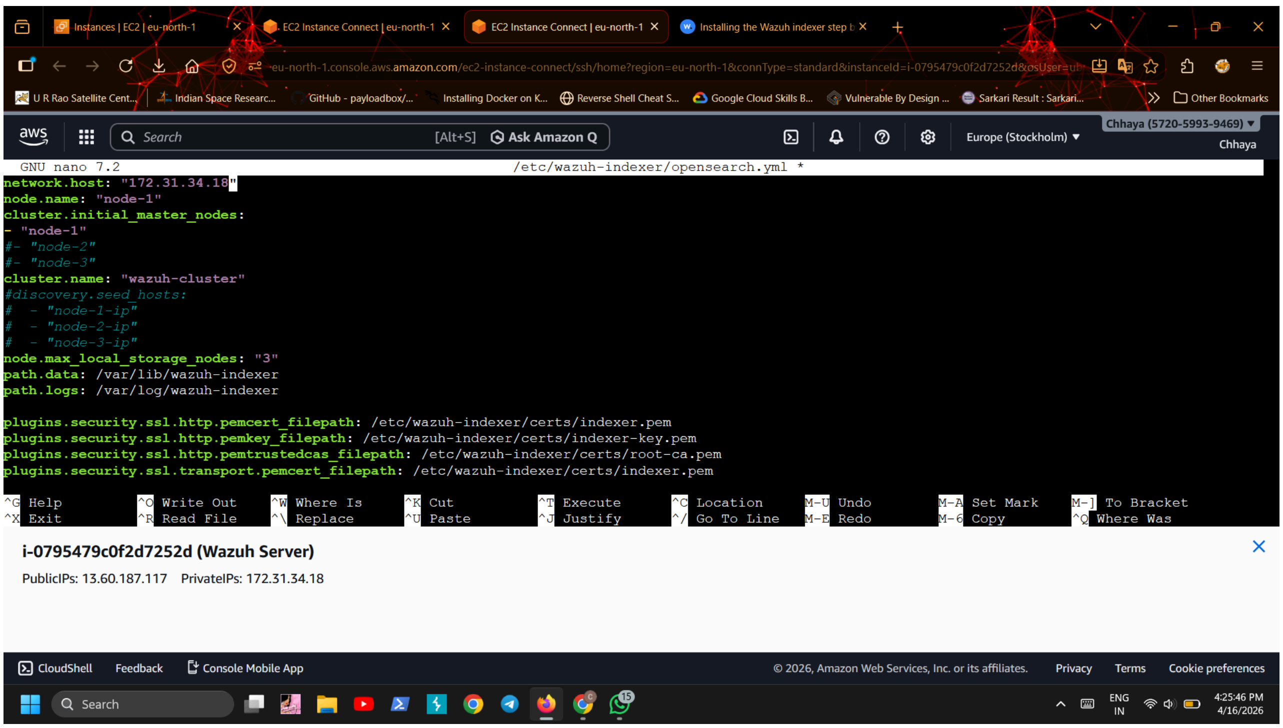
3.3 Indexer Configuration

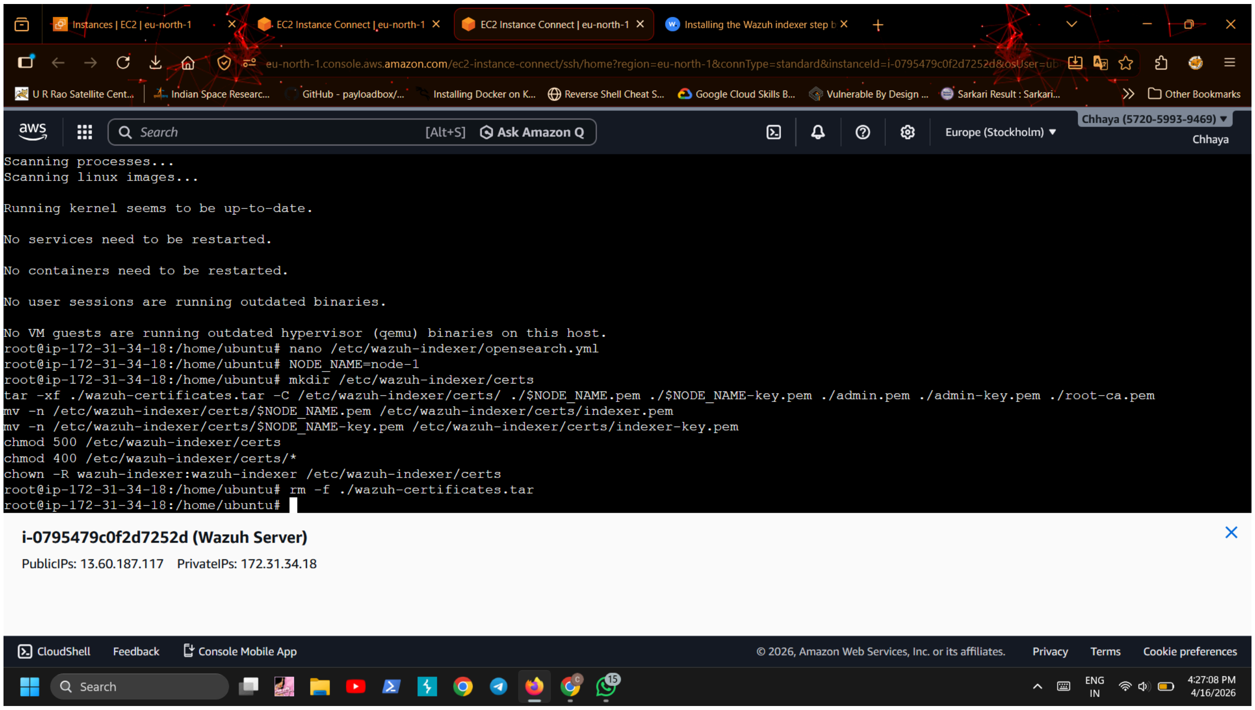
After installation, I configured the indexer:

- Updated network settings (IP address of server)
- Set node name as defined earlier
- Configured cluster settings (single-node in my case)
- Adjusted discovery and node-related configurations



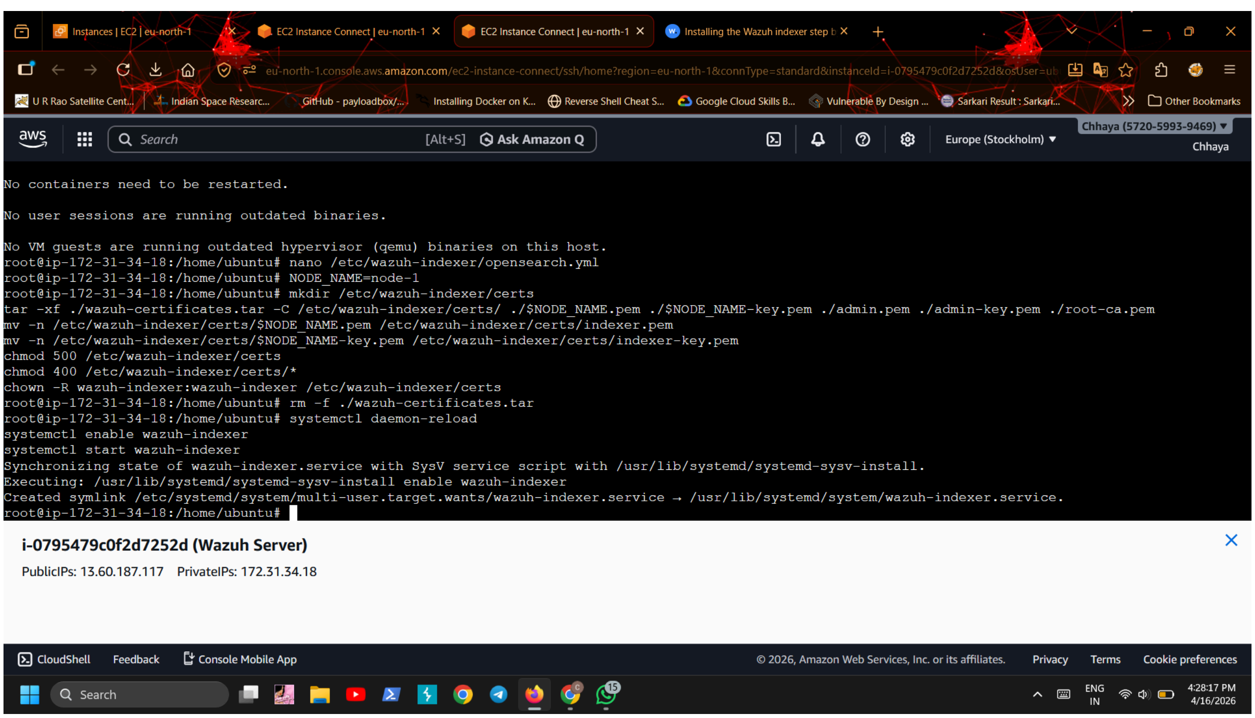
3.4 Deploying Certificates

- Extracted the certificate file
- Moved required certificate files to the correct directory
- Renamed files as required by Wazuh
- Set proper permissions and ownership



3.5 Starting the Indexer Service

- Reloaded system services
- Enabled Wazuh indexer to start automatically
- Started the indexer service

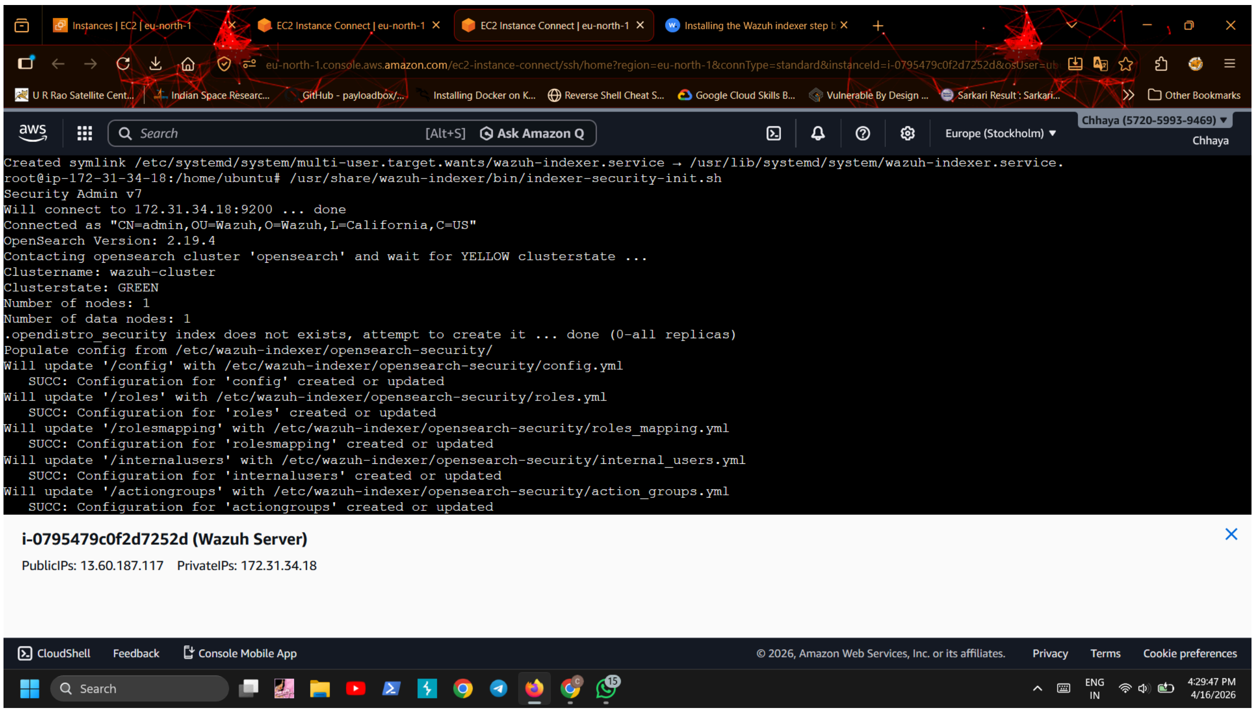


3.6 Cluster Initialization

In the final step:

- Ran the initialization script
- This step loaded certificate information and started the cluster

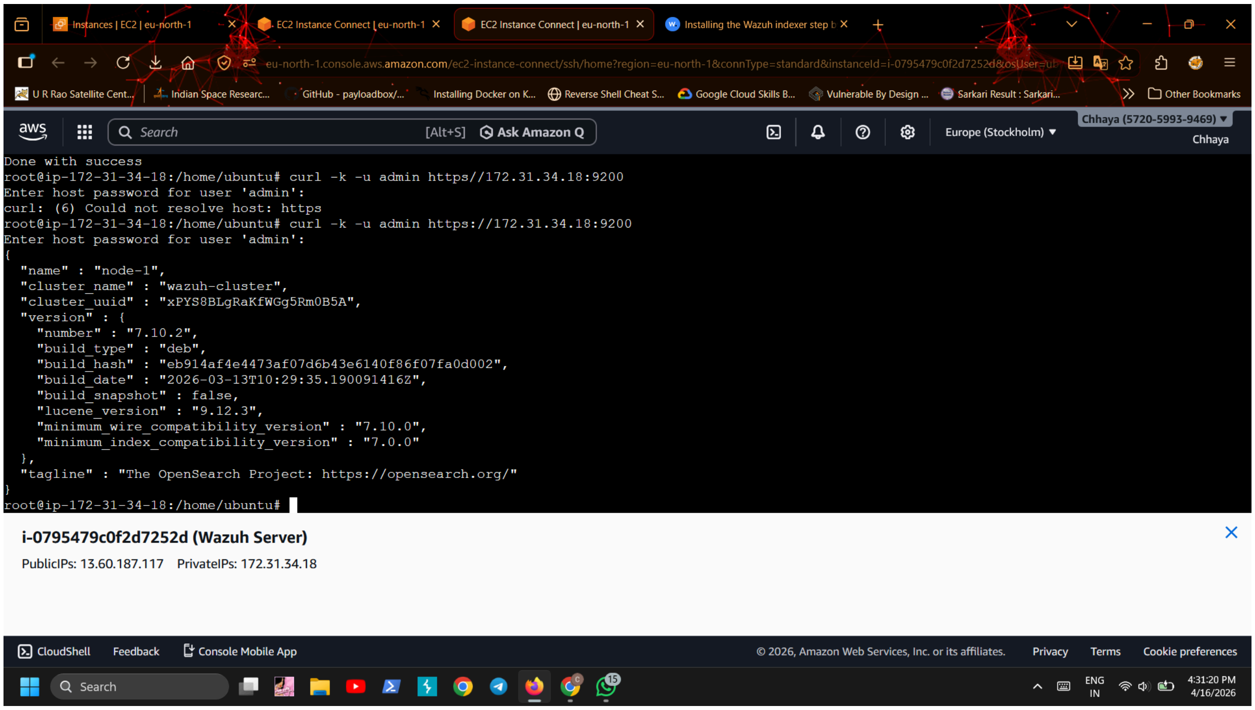
Since I used a single-node setup, this was done only once.

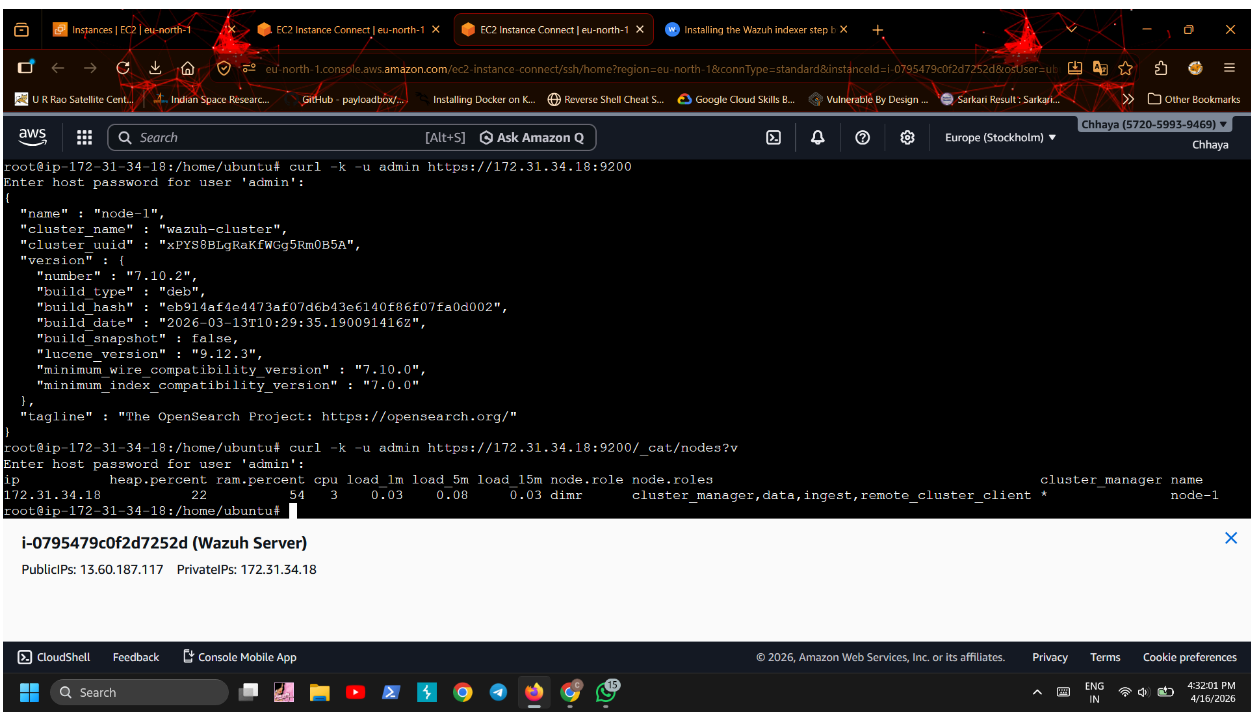


3.7 Testing the Installation

To verify the setup:

- Checked indexer response using API
- Confirmed that the node and cluster were working properly





4. Wazuh Server Installation (Wazuh Server Machine)

After setting up the Wazuh indexer, the next step was to install the Wazuh server on the Wazuh server machine. This component is responsible for collecting logs from agents and generating alerts.

The installation was done in stages as per the official documentation.

4.1 Adding Wazuh Repository

First, I configured the Wazuh repository on the system.

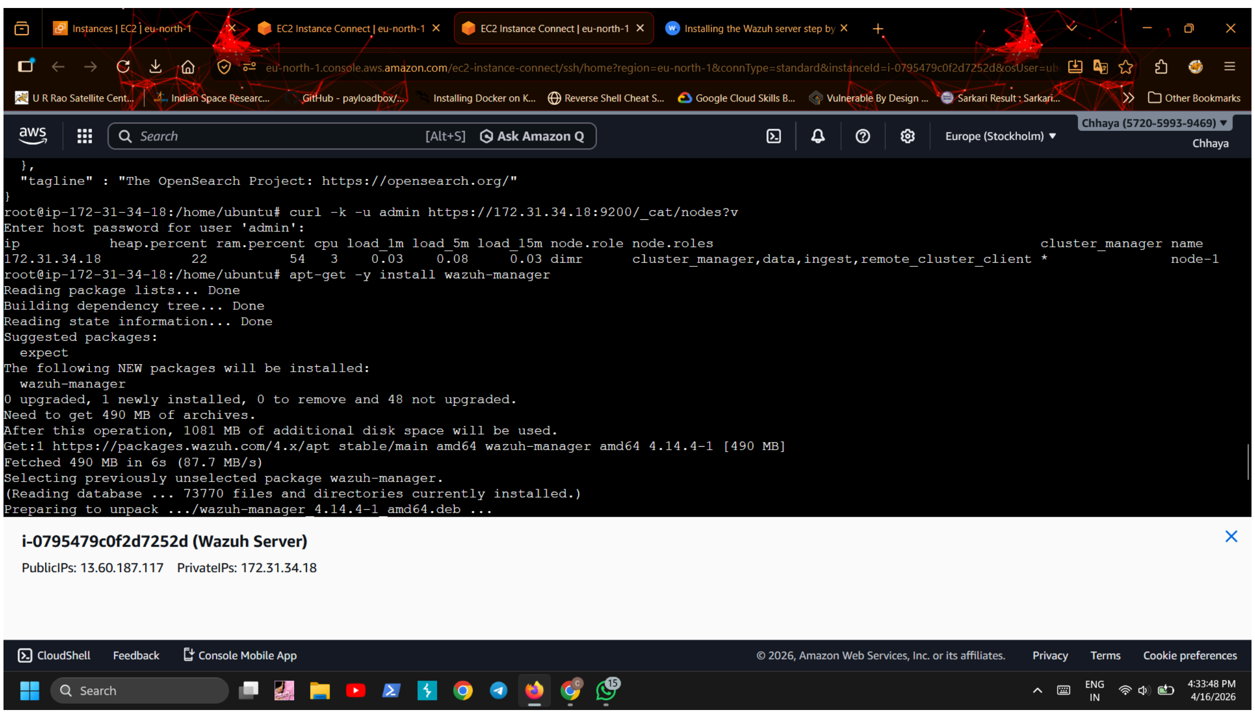
- Installed required packages for secure repository handling
- Added the Wazuh GPG key
- Added the official Wazuh repository
- Updated the system package list

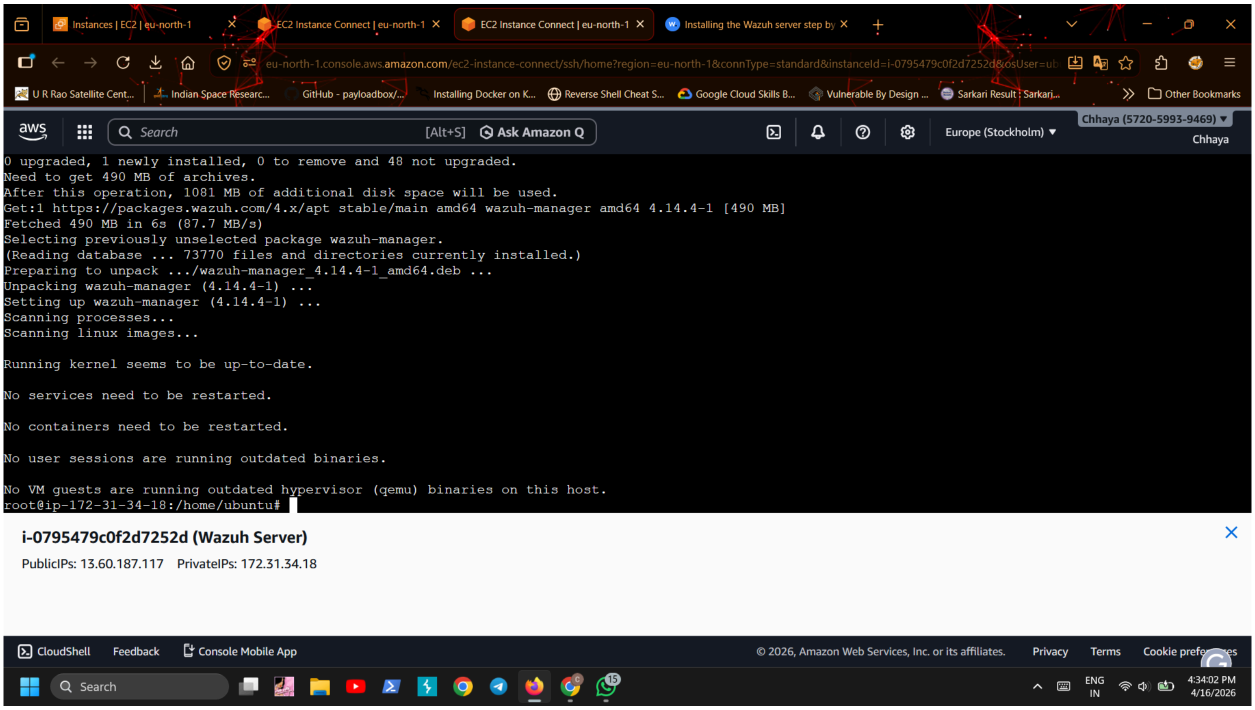
4.2 Installing Wazuh Manager

Next, I installed the Wazuh manager.

- Installed the manager package
- This component is responsible for collecting and analyzing logs from the Wazuh Agent

After installation, the manager was ready for configuration.

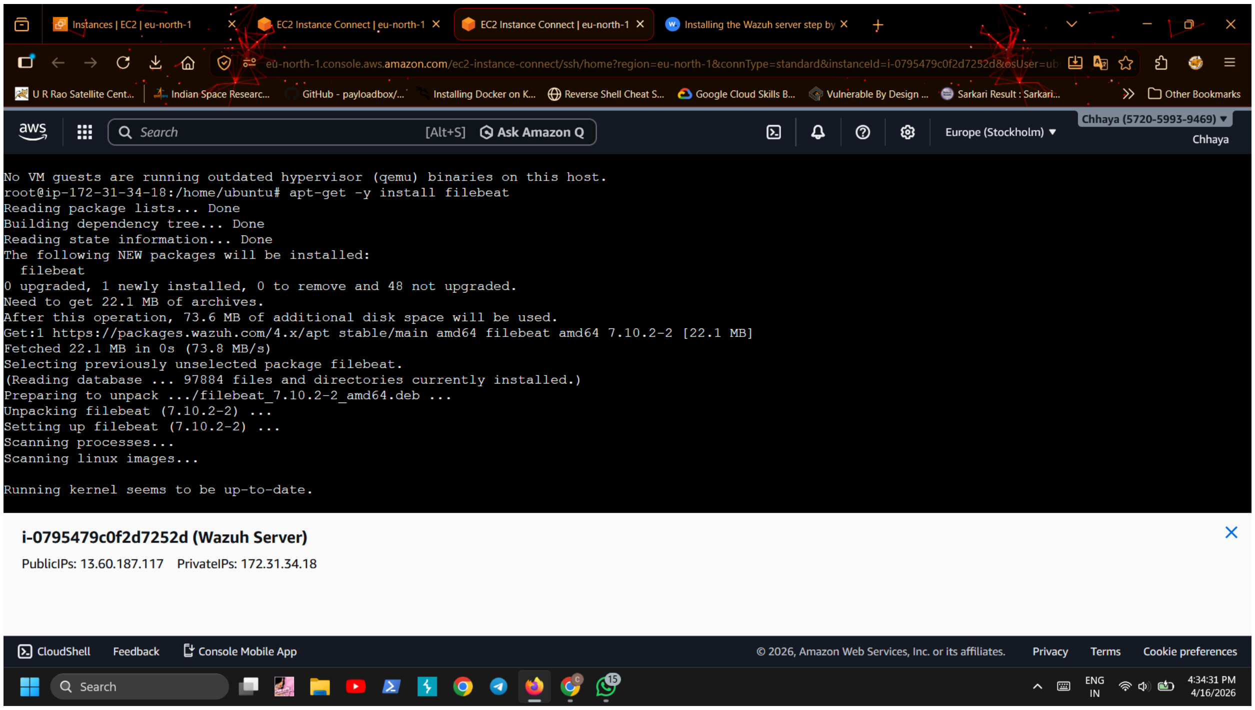




4.3 Installing Filebeat

After installing the manager, I installed Filebeat.

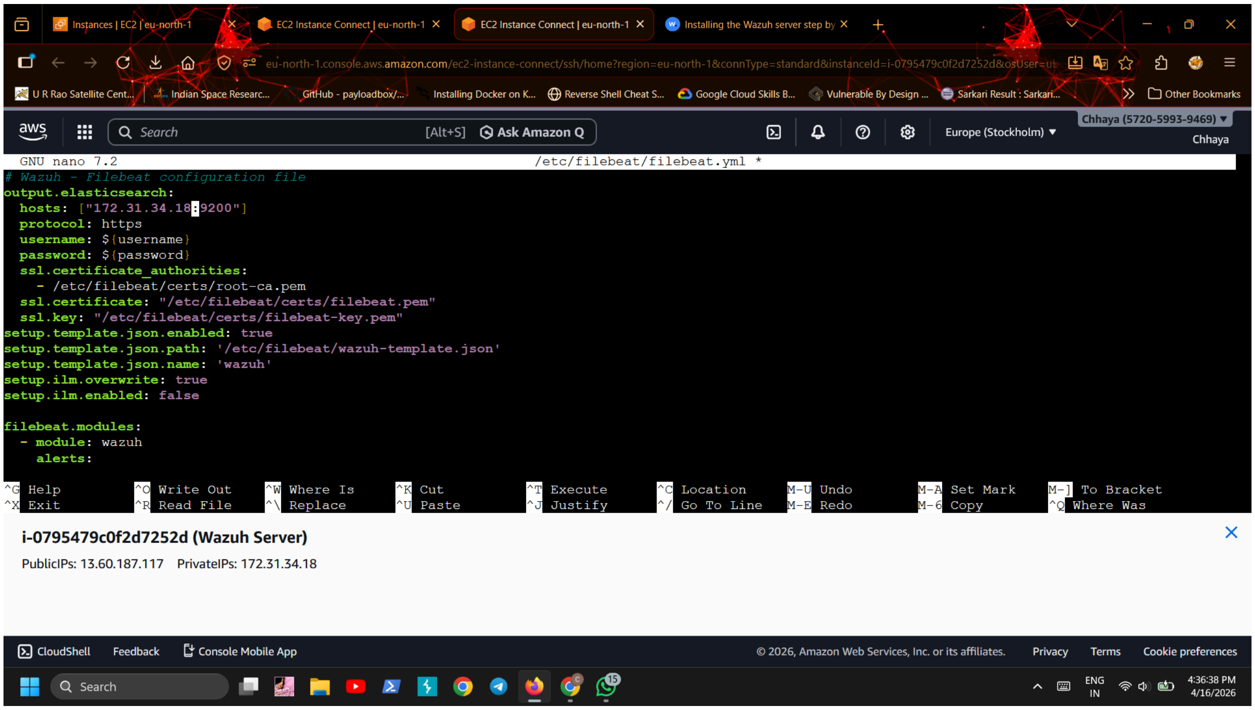
- Filebeat is used to forward alerts and logs to the Wazuh indexer
- It helps in sending data securely from the server to the indexer



4.4 Configuring Filebeat

Then I configured Filebeat to connect with the Wazuh indexer.

- Downloaded the preconfigured Filebeat configuration file
- Updated the indexer IP address (replaced default localhost with actual IP of Wazuh indexer)
- Configured authentication credentials
- Set up secure communication using certificates

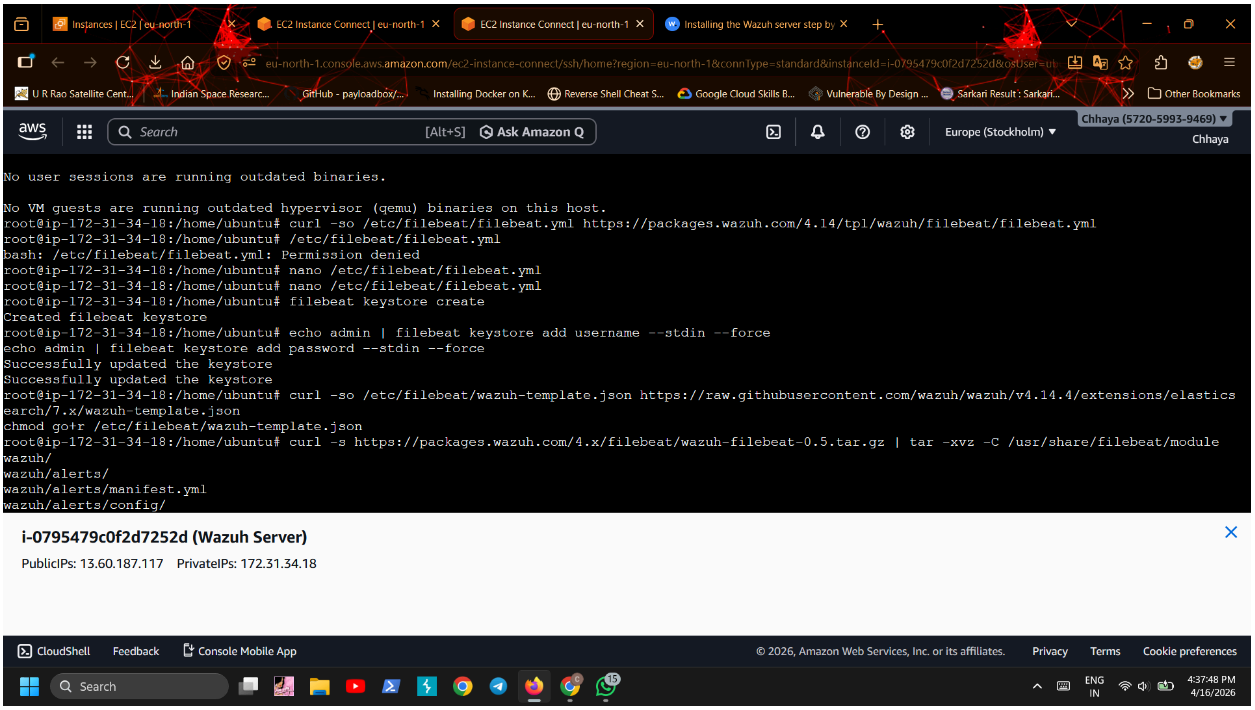


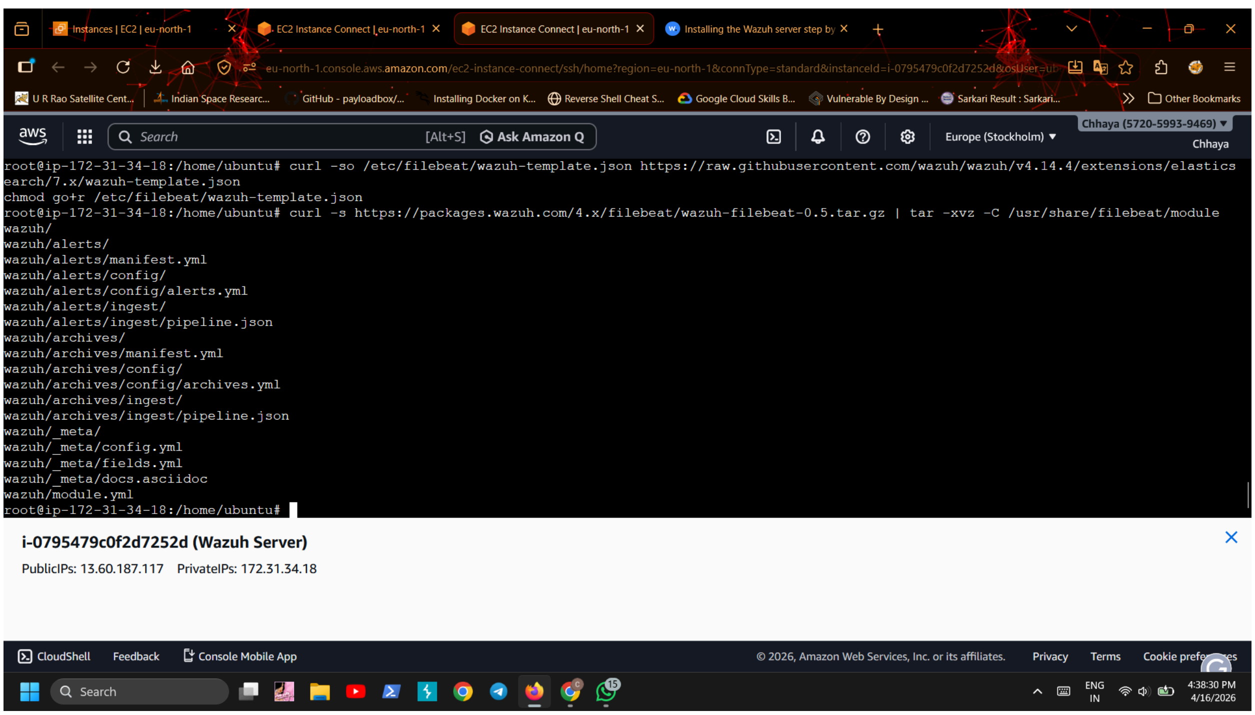
4.5 Setting Up Filebeat Keystore

To store credentials securely:

- Created Filebeat keystore
- Added username and password (default admin credentials)

This ensures sensitive data is not stored in plain text.

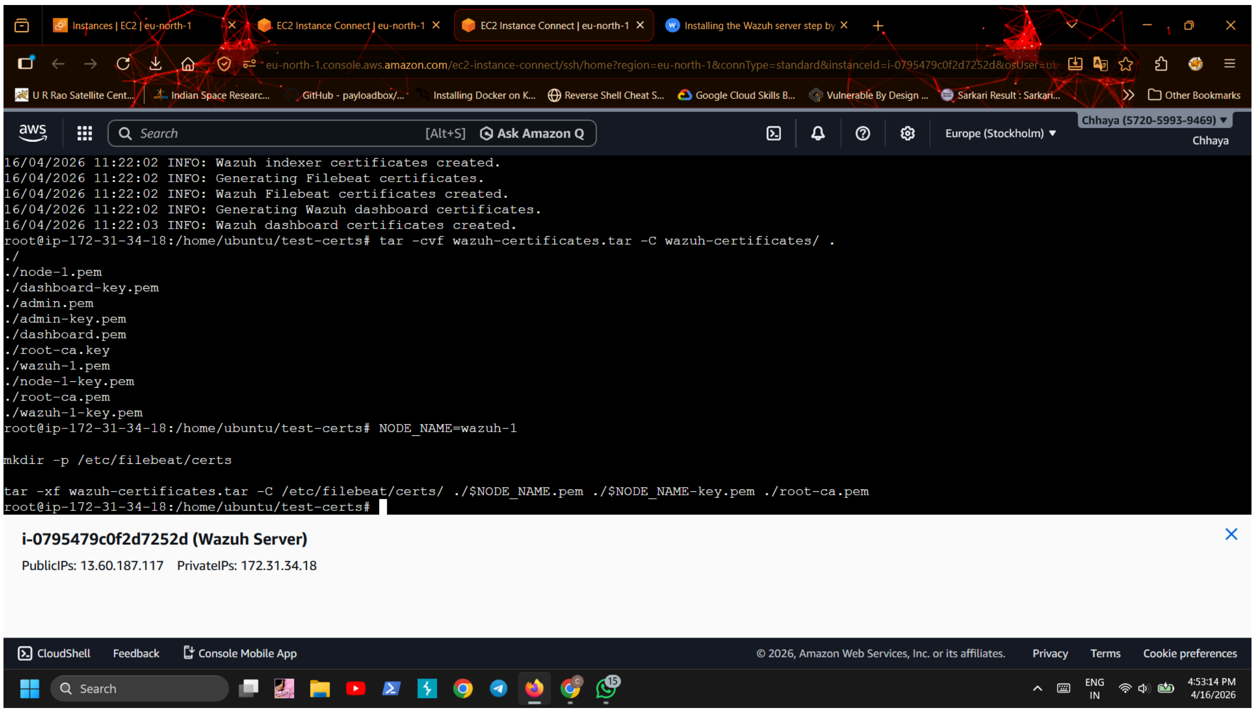

No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-34-18:/home/ubuntu# curl -so /etc/filebeat/filebeat.yml https://packages.wazuh.com/4.14/tpl/wazuh/filebeat/filebeat.yml
root@ip-172-31-34-18:/home/ubuntu# /etc/filebeat/filebeat.yml
bash: /etc/filebeat/filebeat.yml: Permission denied
root@ip-172-31-34-18:/home/ubuntu# nano /etc/filebeat/filebeat.yml
root@ip-172-31-34-18:/home/ubuntu# nano /etc/filebeat/filebeat.yml
root@ip-172-31-34-18:/home/ubuntu# filebeat keystore create
Created filebeat keystore
root@ip-172-31-34-18:/home/ubuntu# echo admin | filebeat keystore add username --stdin --force
echo admin | filebeat keystore add password --stdin --force
Successfully updated the keystore
Successfully updated the keystore
root@ip-172-31-34-18:/home/ubuntu# curl -so /etc/filebeat/wazuh-template.json https://raw.githubusercontent.com/wazuh/wazuh/v4.14.4/extensions/elasticsearch/7.x/wazuh-template.json
chmod go+r /etc/filebeat/wazuh-template.json
root@ip-172-31-34-18:/home/ubuntu# curl -s https://packages.wazuh.com/4.x/filebeat/wazuh-filebeat-0.5.tar.gz | tar -xvz -C /usr/share/filebeat/module
wazuh/
wazuh/alerts/
wazuh/alerts/manifest.yml
wazuh/alerts/config/
i-0795479c0f2d7252d (Wazuh Server)
PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18


root@ip-172-31-34-18:/home/ubuntu# curl -so /etc/filebeat/wazuh-template.json https://raw.githubusercontent.com/wazuh/wazuh/v4.14.4/extensions/elasticsearch/7.x/wazuh-template.json
chmod go+r /etc/filebeat/wazuh-template.json
root@ip-172-31-34-18:/home/ubuntu# curl -s https://packages.wazuh.com/4.x/filebeat/wazuh-filebeat-0.5.tar.gz | tar -xvz -C /usr/share/filebeat/module
wazuh/
wazuh/alerts/
wazuh/alerts/manifest.yml
wazuh/alerts/config/
wazuh/alerts/config/alerts.yml
wazuh/alerts/ingest/
wazuh/alerts/ingest/pipeline.json
wazuh/archives/
wazuh/archives/manifest.yml
wazuh/archives/config/
wazuh/archives/config/archives.yml
wazuh/archives/ingest/
wazuh/archives/ingest/pipeline.json
wazuh/_meta/
wazuh/_meta/config.yml
wazuh/_meta/fields.yml
wazuh/_meta/docs.asciidoc
wazuh/module.yml
root@ip-172-31-34-18:/home/ubuntu#
i-0795479c0f2d7252d (Wazuh Server)
PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18

4.6 Deploying Certificates

After that, I deployed certificates for secure communication.

- Extracted certificates generated earlier
- Placed them in the correct Filebeat directory
- Renamed files as required
- Set proper permissions



4.7 Configuring Indexer Connection in Wazuh Server

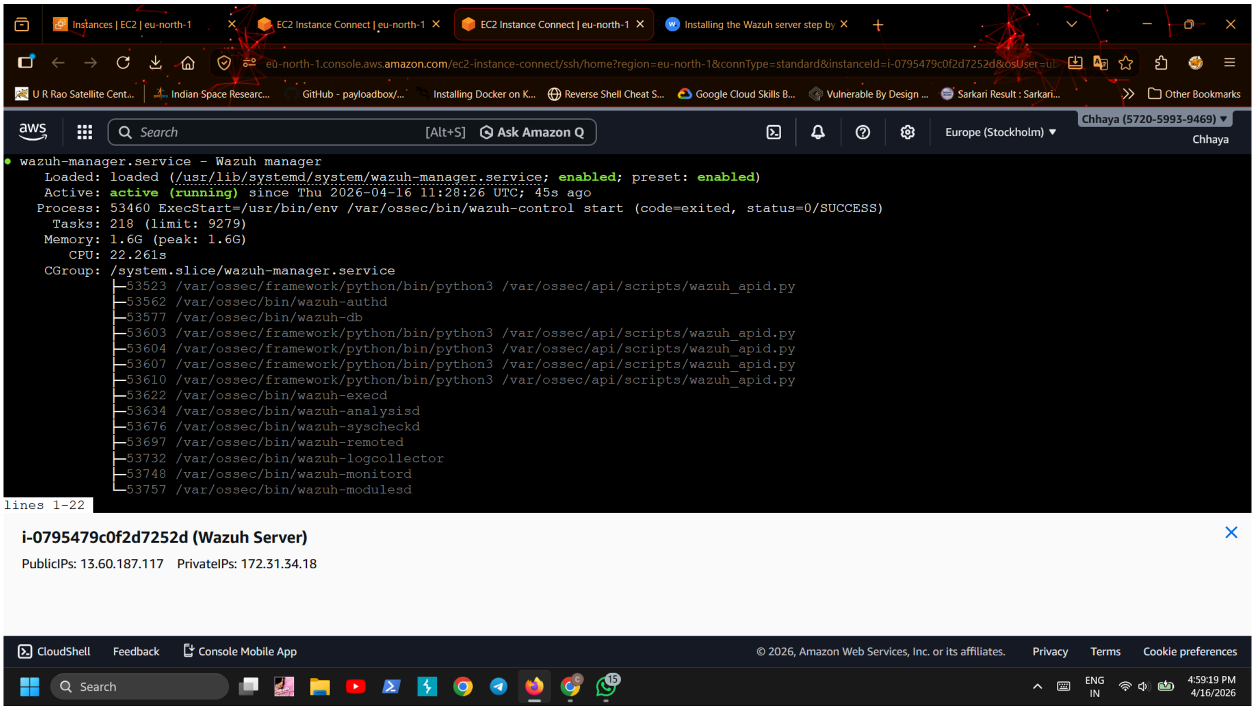
Next, I configured the connection between Wazuh server and indexer.

- Stored indexer credentials in Wazuh keystore
- Updated configuration file to include indexer IP address
- Verified certificate paths

4.8 Starting Wazuh Manager Service

After configuration:

- Enabled and started Wazuh manager service
- Verified that the service was running properly

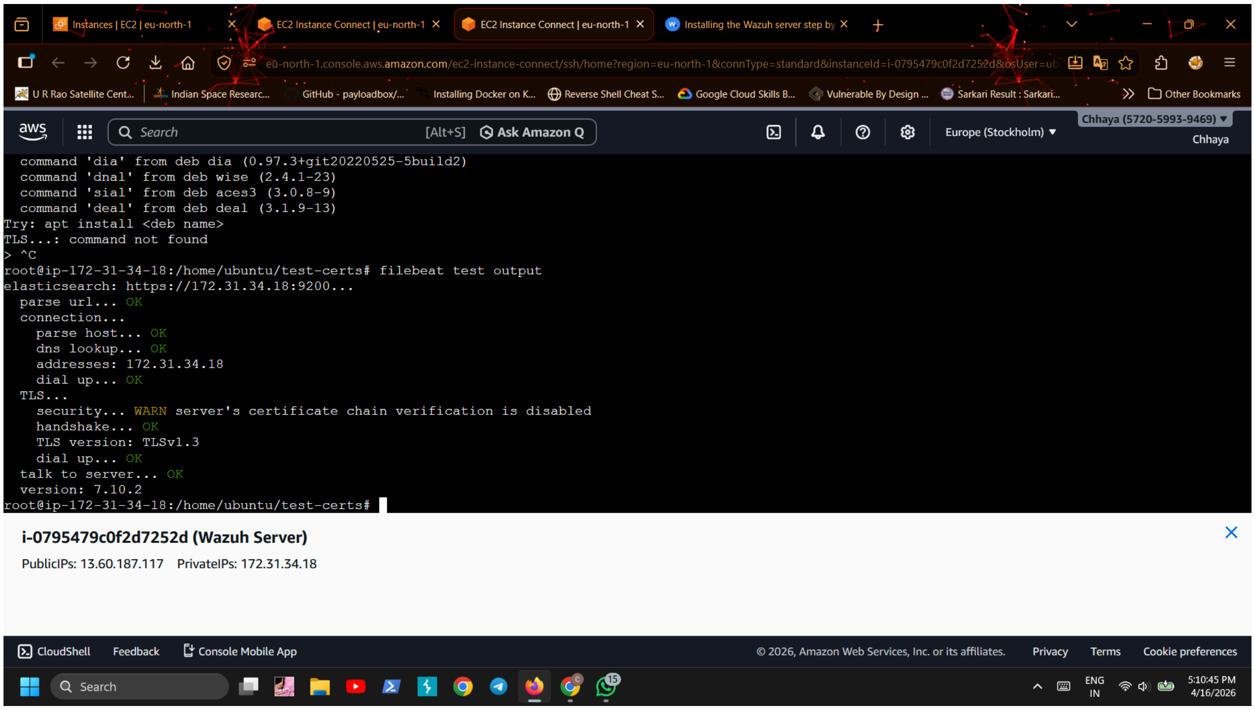


4.9 Starting Filebeat Service

Then I started Filebeat:

- Enabled and started Filebeat service
- Tested the connection to ensure logs are being forwarded correctly

The output confirmed successful communication with the indexer.

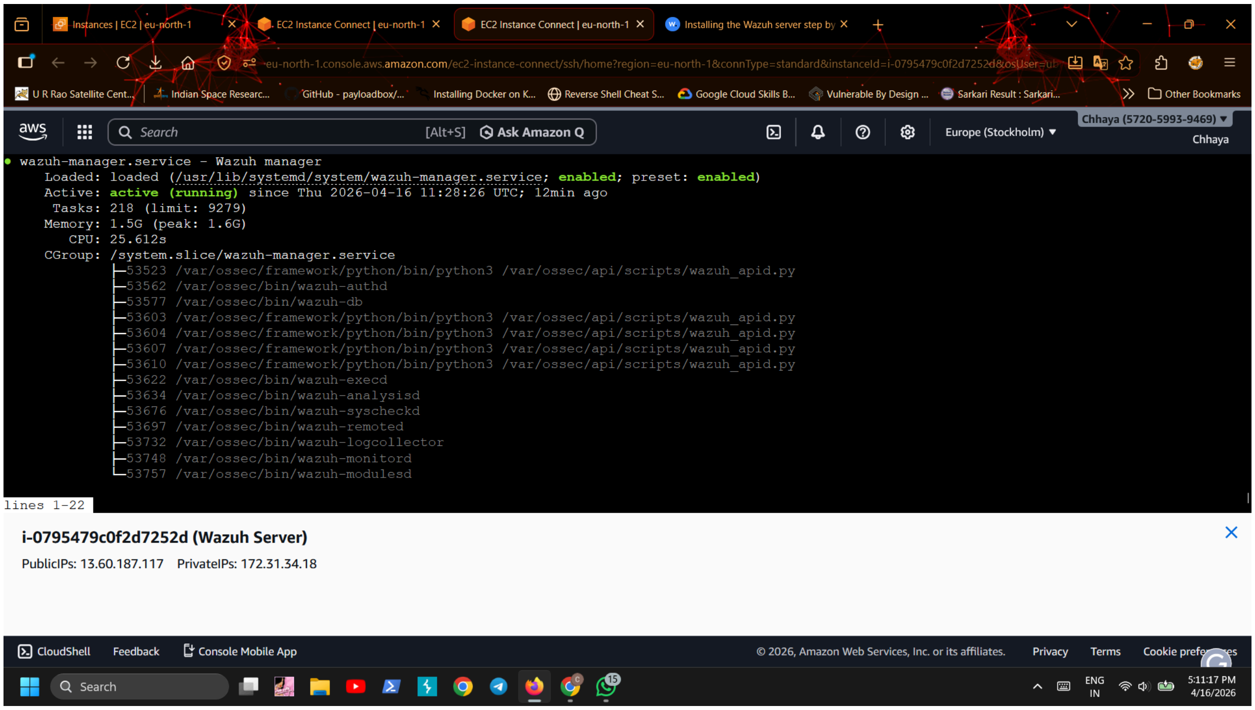


4.10 Verification

After completing all steps:

- Wazuh manager was running successfully
- Filebeat was forwarding logs to indexer
- No errors were observed in service status

This confirmed that the Wazuh server setup was completed successfully.



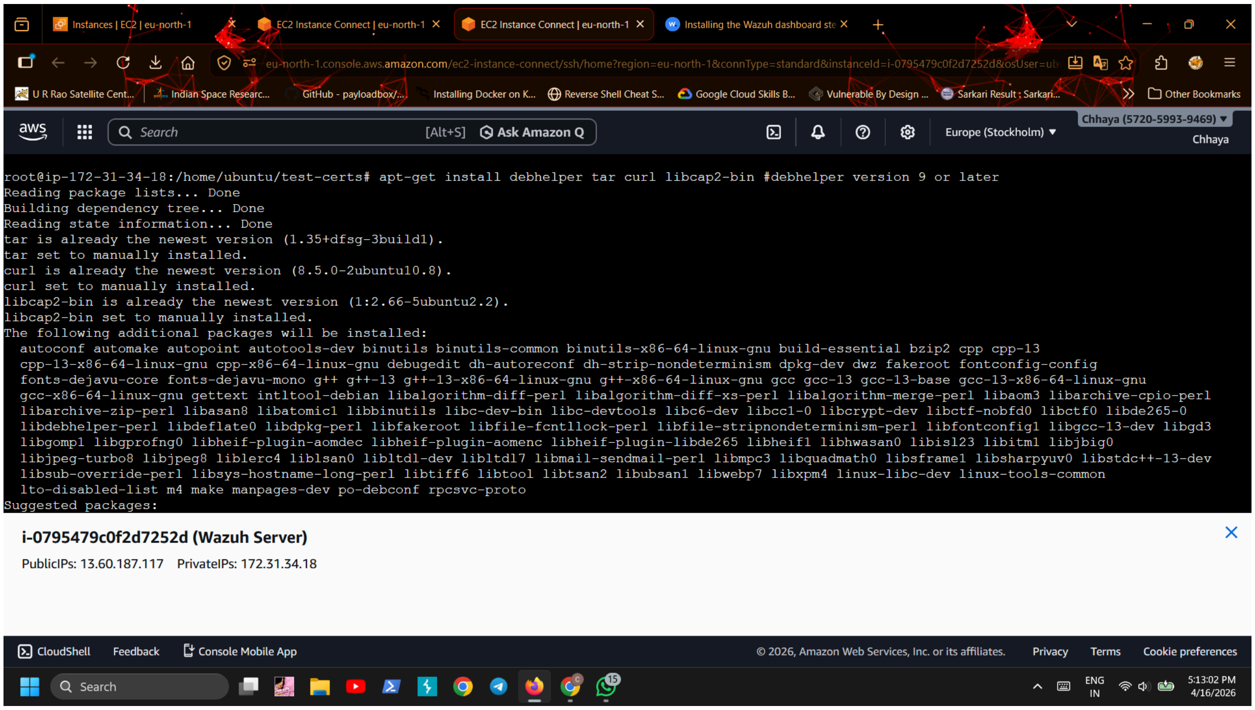
5. Wazuh Dashboard Installation (Wazuh Server Machine)

After completing the installation of the Wazuh indexer and Wazuh server, the final component installed was the Wazuh Dashboard. This is the web interface used to visualize alerts, monitor agents, and analyze security events.

5.1 Installing Required Dependencies

First, I installed the required system packages needed for the dashboard installation.

These packages help in handling archives, downloads, and system utilities required during setup.



5.2 Adding Wazuh Repository

Next, I configured the Wazuh repository on the system.

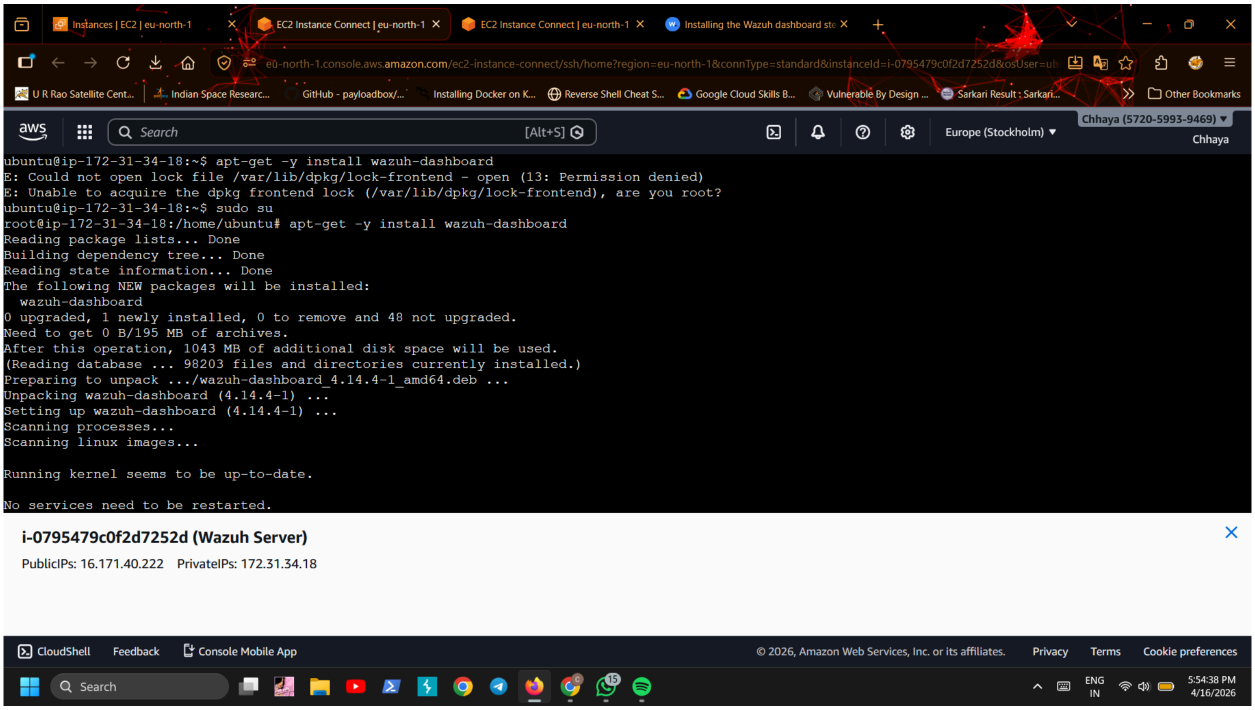
- Installed required repository tools
- Added the Wazuh GPG key for secure package verification
- Added the official Wazuh repository
- Updated system package list

This ensures that the latest Wazuh dashboard package can be installed.

5.3 Installing Wazuh Dashboard

After repository setup:

- Installed the Wazuh dashboard package
- This provides the web-based interface for monitoring logs and alerts

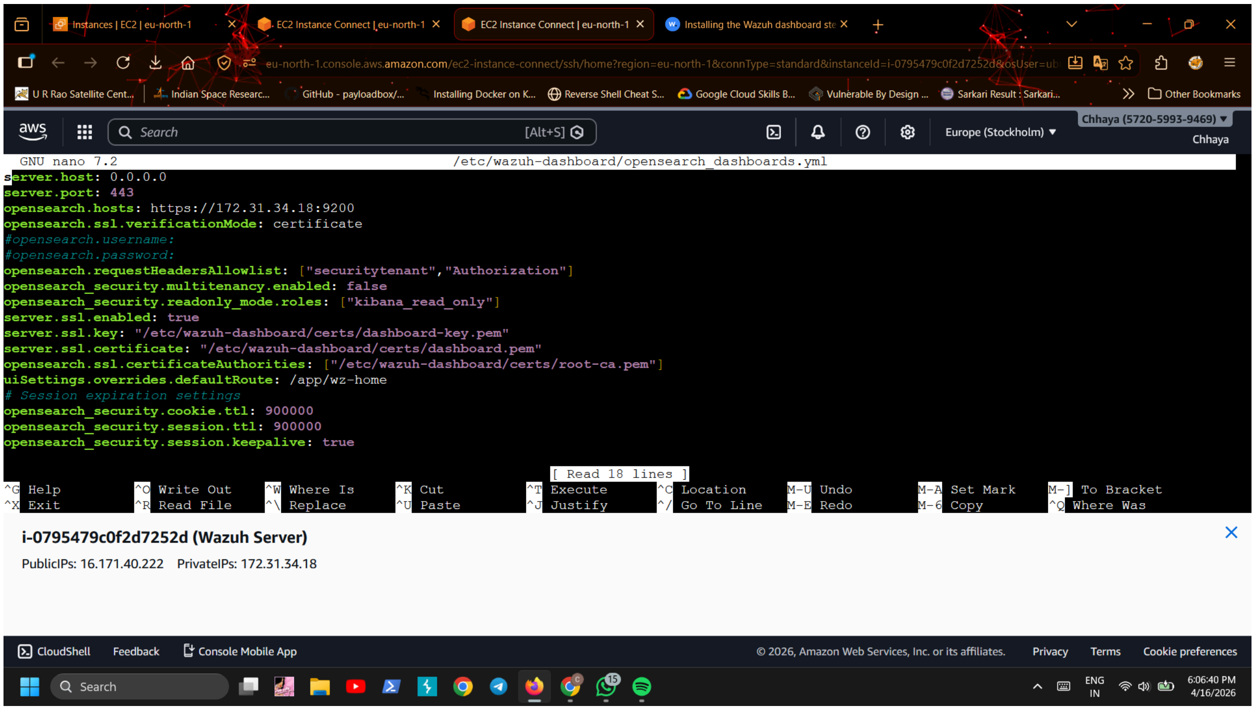


5.4 Configuring Wazuh Dashboard

After installation, I configured the dashboard settings.

- Set dashboard to listen on all available network interfaces
- Configured port settings for secure access
- Added Wazuh indexer connection details
- Disabled SSL verification mode for local setup (as per lab environment)

This step ensures the dashboard can communicate with the indexer properly.

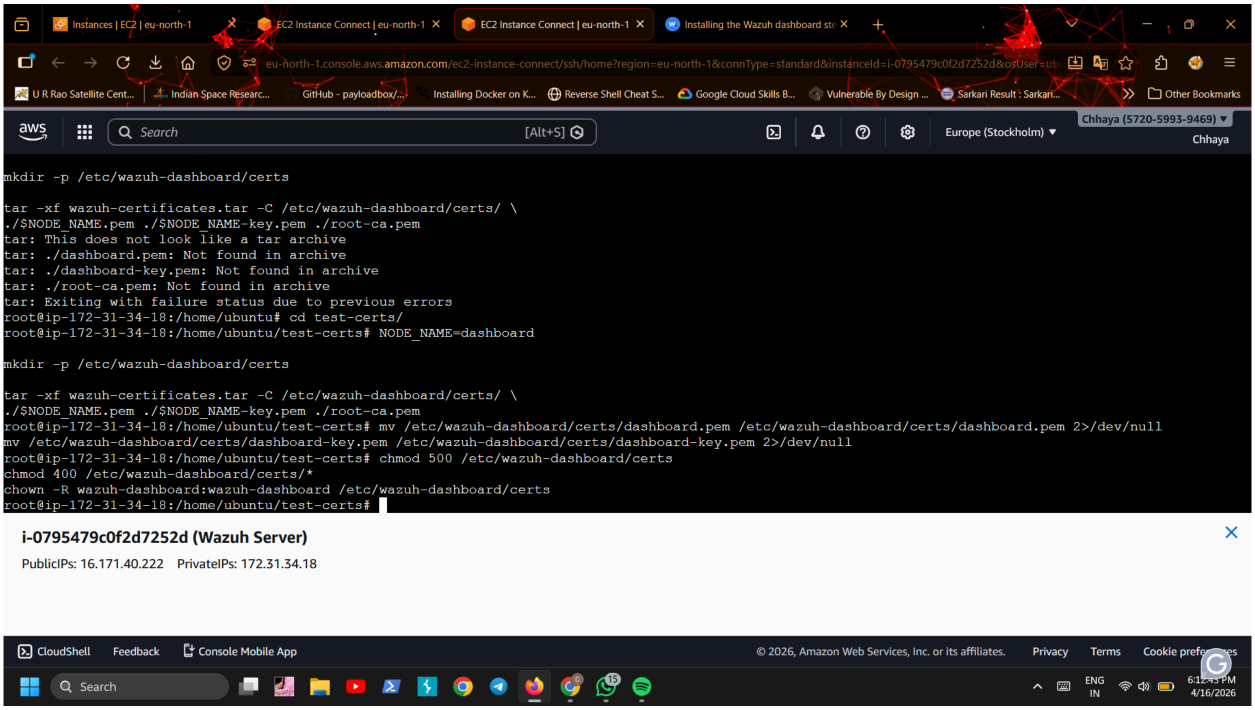


5.5 Deploying Certificates

Next, I deployed SSL certificates for secure communication.

- Extracted certificates generated earlier during installation
- Copied them into the Wazuh dashboard certificate directory
- Renamed certificate files as required
- Set proper permissions and ownership

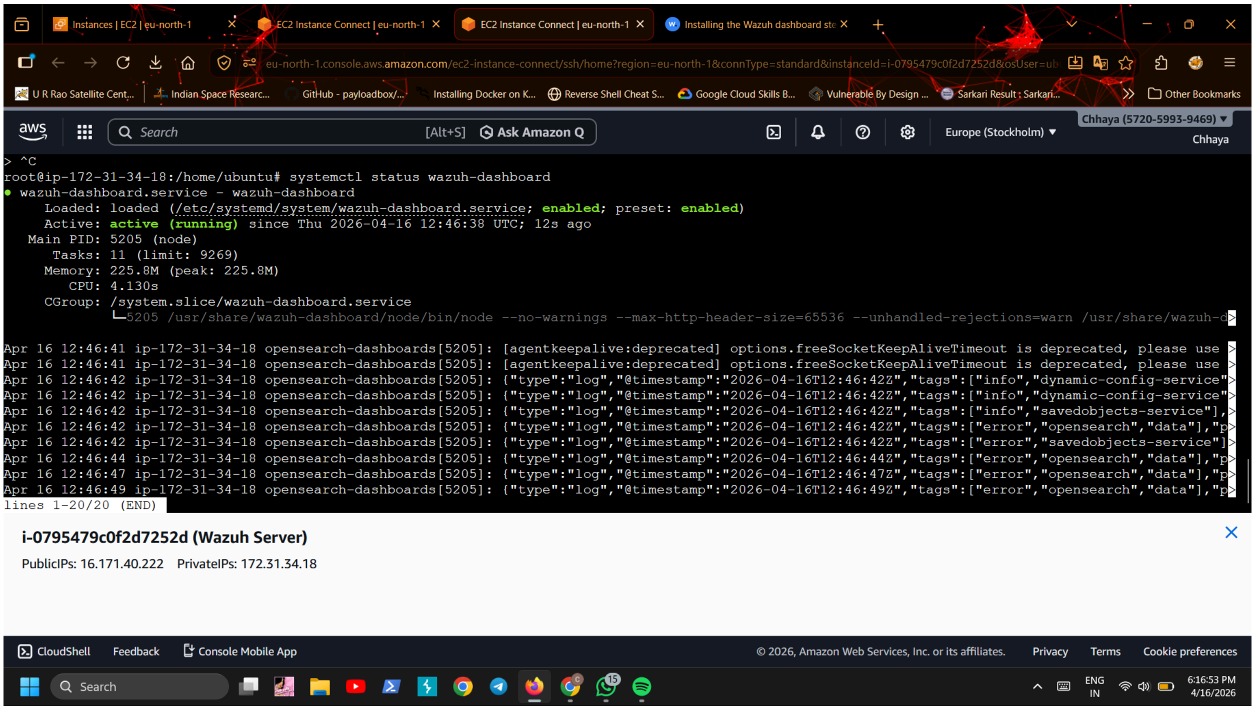
This ensures encrypted communication between the dashboard and other components.



5.6 Starting Wazuh Dashboard Service

After configuration:

- Enabled Wazuh dashboard service
- Started the service using system controls
- Verified that the dashboard was running successfully



5.7 Connecting Dashboard to Wazuh Server

Then I configured the connection between the dashboard the Wazuh server:

- Updated Wazuh server IP in dashboard configuration file
- Used default API credentials for connection
- Enabled secure communication between dashboard and server

This allows the dashboard to fetch alerts and logs from the Wazuh server.

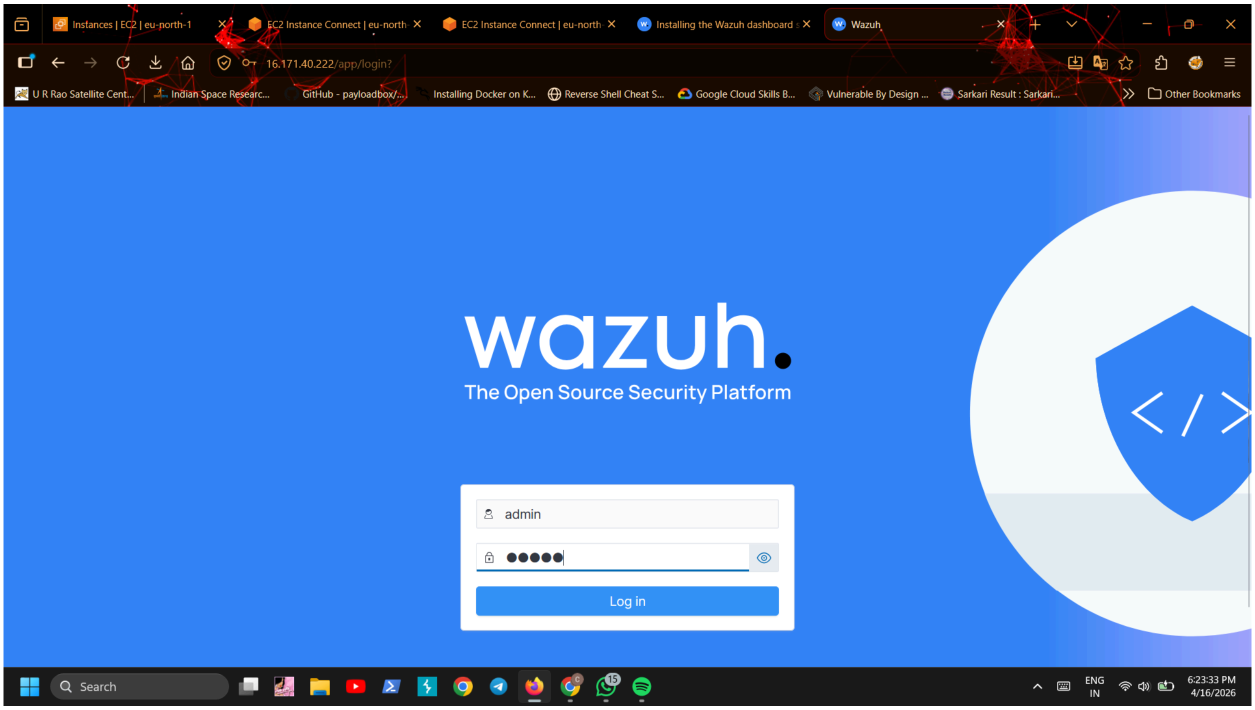
5.8 Accessing Wazuh Dashboard

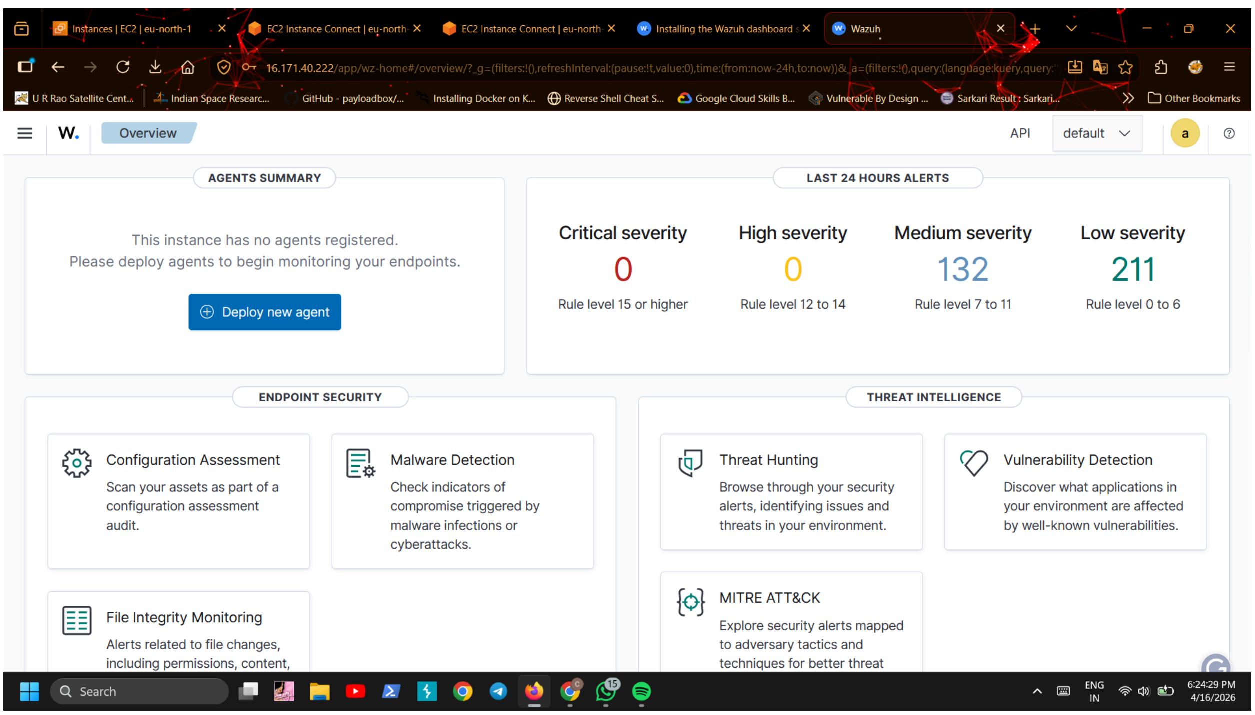
After setup, I accessed the Wazuh web interface:

- Opened dashboard using browser
- Used default credentials:
    - Username: admin
    - Password: admin

On first login, the browser showed a certificate warning due to self-signed certificates. This was expected in a local lab setup and was accepted manually.

After login, the Wazuh dashboard loaded successfully.





5.9 Disabling Updates (Optional Step)

After completing installation, I disabled automatic repository updates to avoid accidental upgrades that could break the setup.

5.10 Securing Installation (Important Step)

Finally, I updated default credentials for security purposes.

- Changed internal user passwords using Wazuh security tool
- Updated API and indexer credentials
- Restarted services to apply changes

This step improves security of the entire Wazuh environment.

6. Wazuh Agent Installation (Wazuh Agent Machine)

After completing the setup of the Wazuh Server, Indexer, and Dashboard, the next step was to install the Wazuh Agent on a separate virtual machine named Wazuh Agent.

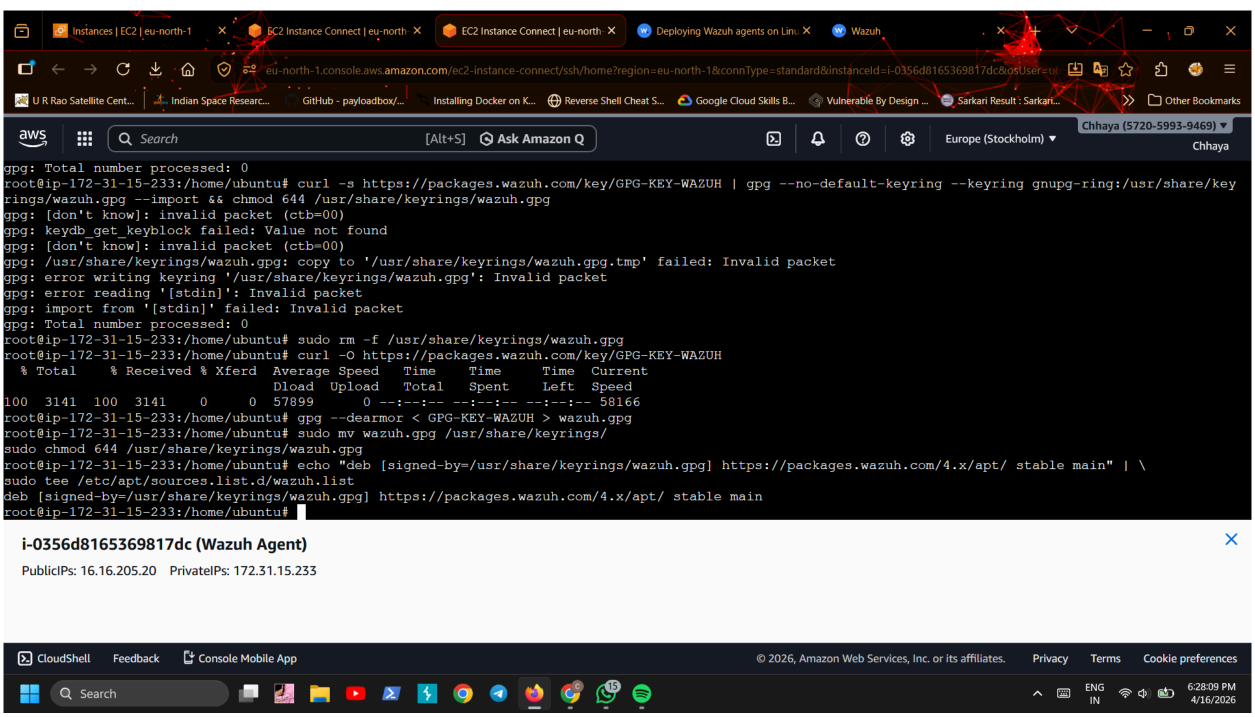
The Wazuh Agent is responsible for collecting logs and system activity from the endpoint and sending them securely to the Wazuh Server for analysis.

6.1 Adding Wazuh Repository

First, I added the official Wazuh repository on the Wazuh Agent machine.

- Installed required packages for repository handling
- Added the Wazuh GPG key for secure package verification
- Added Wazuh repository to the system
- Updated package list

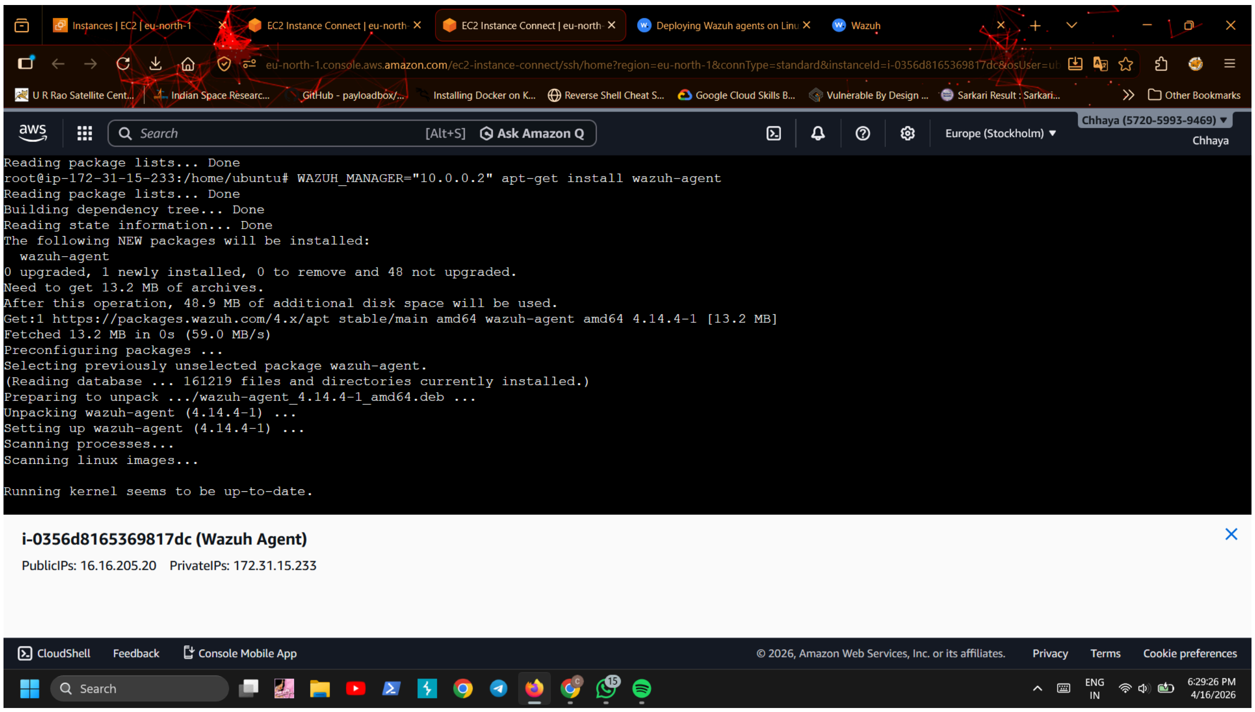
This step ensures the system can download the official Wazuh agent package.



6.2 Installing Wazuh Agent

After adding the repository, I installed the Wazuh agent.

- Installed the Wazuh agent package on the system
- While installing, I provided the **Wazuh Server IP address** so that the agent can automatically connect to the server
- This step links the agent with the central SIEM system

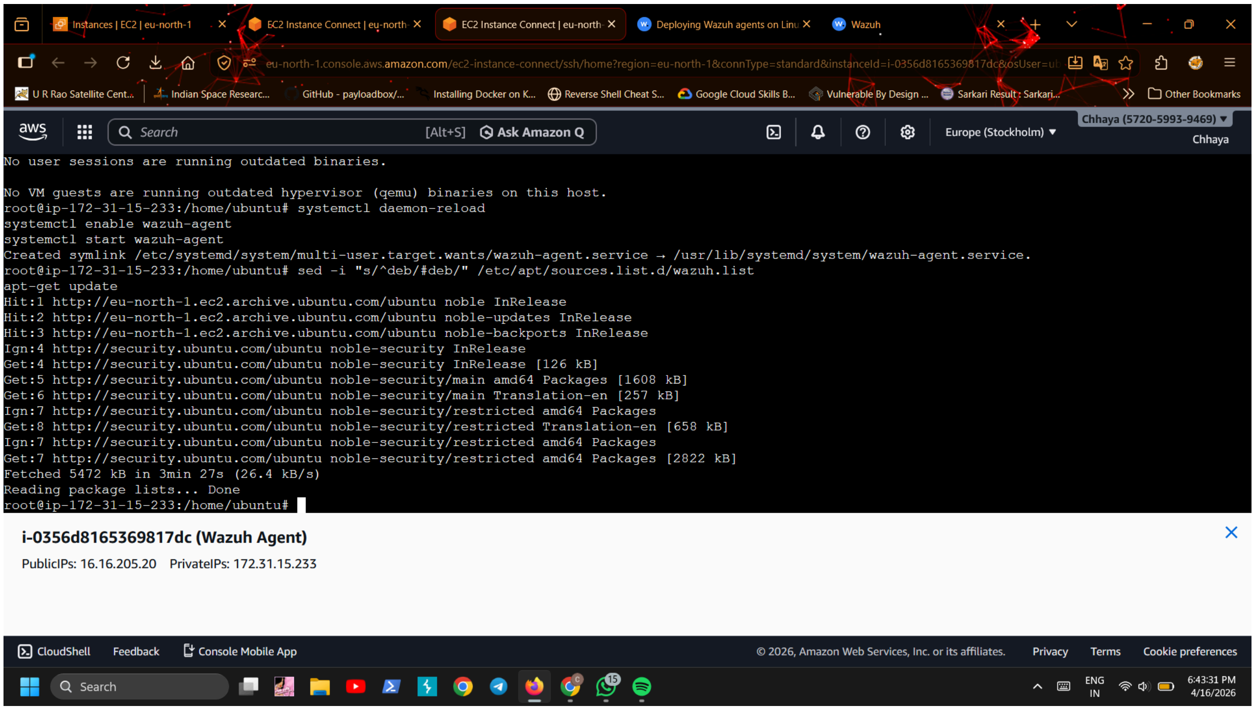


6.3 Starting Wazuh Agent Service

After installation:

- Reloaded system services
- Enabled Wazuh agent to start automatically on boot
- Started the Wazuh agent service

Once started, the agent began communicating with the Wazuh Server.

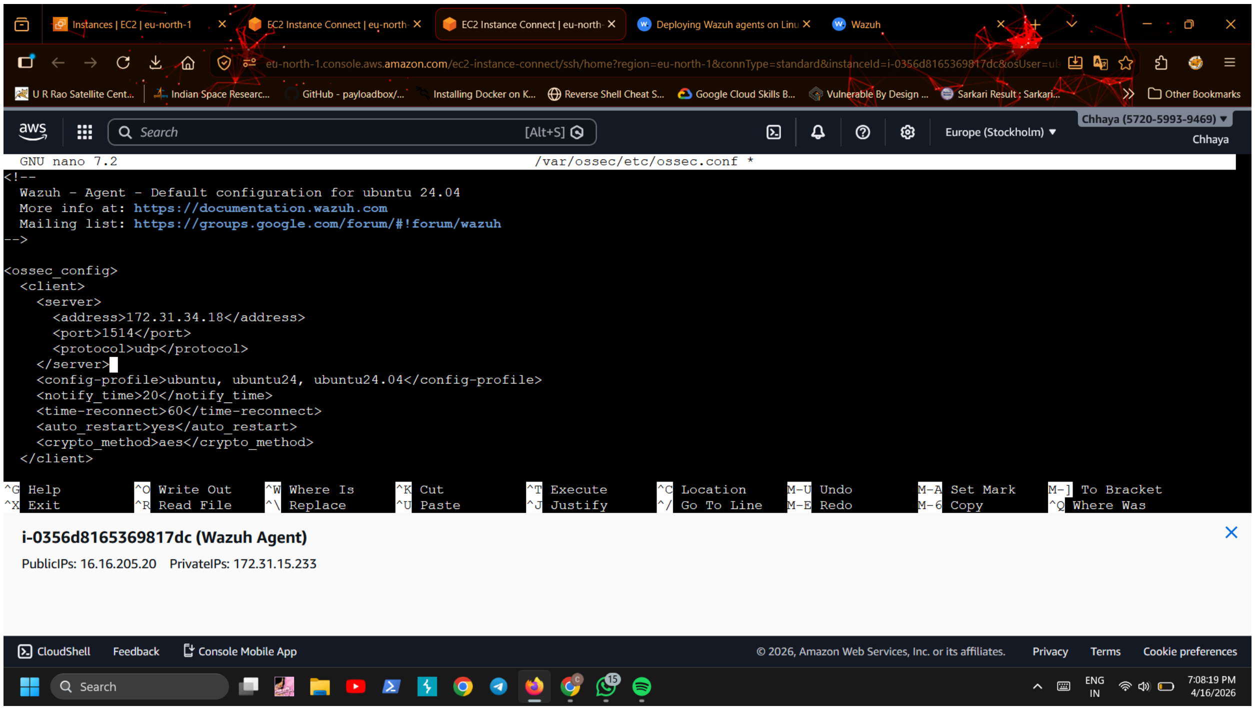


6.4 Verifying Agent Connection

After starting the service:

- Checked the Wazuh dashboard on the **Wazuh Server**
- Verified that the **Wazuh Agent** was listed as active
- Confirmed successful communication between agent and server

This confirmed that log forwarding was working correctly.



6.5 Disabling Repository Updates (Optional Step)

After successful installation, I disabled automatic updates for Wazuh packages to avoid accidental version mismatches.

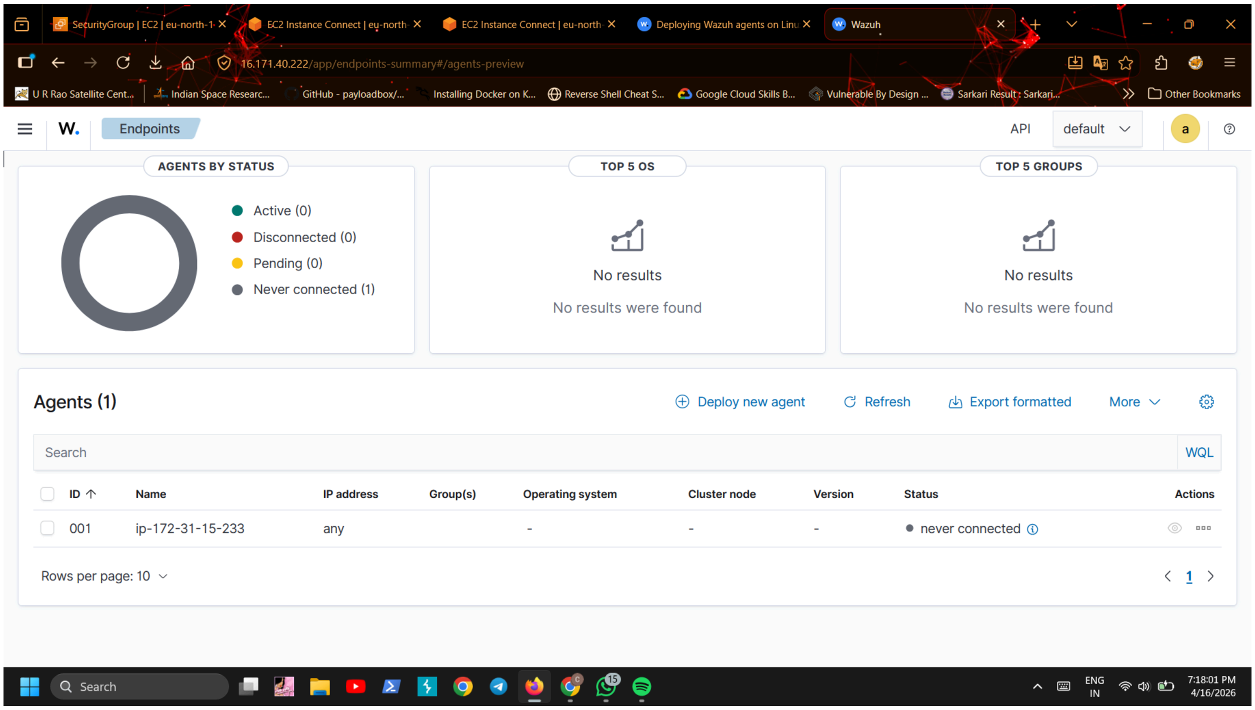
This ensures compatibility between the Wazuh Agent and Wazuh Server.

6.6 Final Status Check

At the end of the setup:

- Wazuh Agent was installed successfully
- It was connected to the Wazuh Server
- Logs from the agent were being received in the dashboard

This confirmed that the endpoint monitoring setup was working correctly.



Detection & Prevention Lab

1. Objective

* Detect network scans (e.g., Nmap)

* Log firewall activity

* Send logs to Wazuh SIEM

* Trigger alerts

* (Optional) Block attacker automatically

2. Architecture

text

Attacker → Agent (UFW/iptables logs)

→ Wazuh Agent collects logs

→ Wazuh Manager analyzes logs

→ Alerts generated in Dashboard

→ Active response blocks attacker (optional)

3. Environment

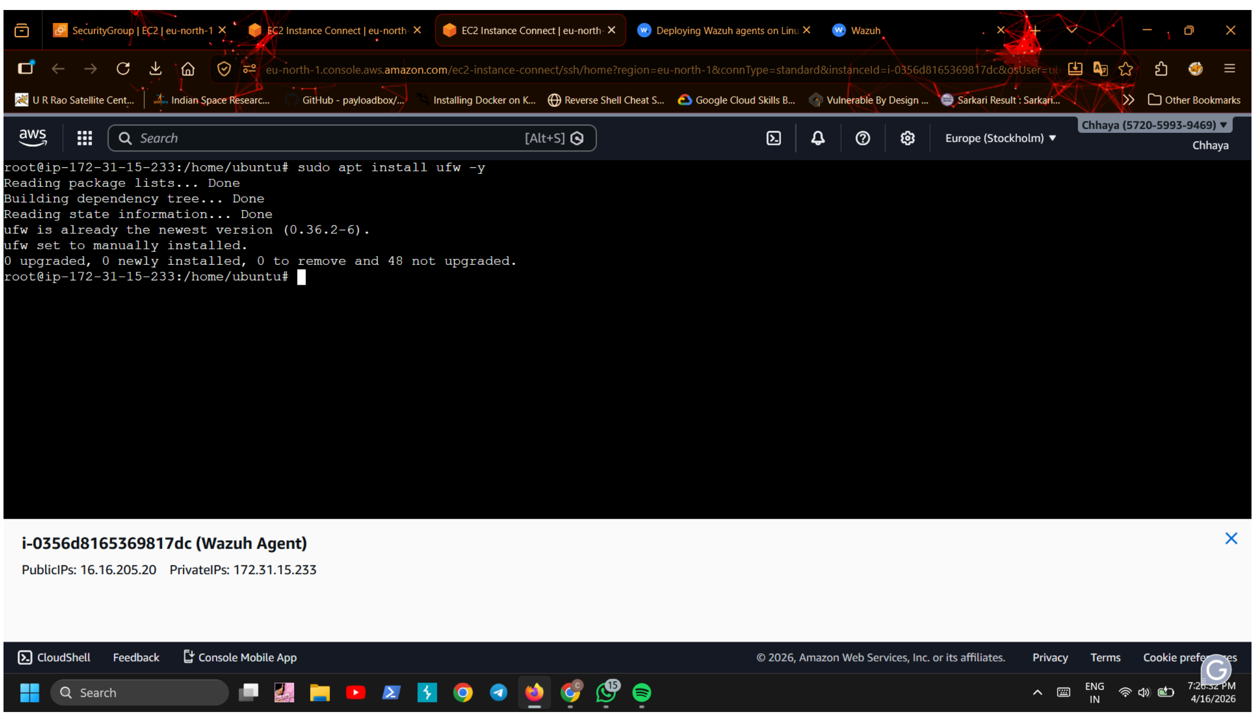
Component	IP
Wazuh Manager	172.31.34.225
Wazuh Agent	172.31.2.238
OS	Ubuntu
Firewall	UFW + iptables

4. Firewall Configuration (Agent)

4.1 Install UFW

bash

```
sudo apt install ufw -y
```



4.2 Allow Required Ports

bash

```
sudo ufw allow 22/tcp      # SSH (IMPORTANT)
```

sudo ufw allow 1514/udp   # Wazuh logs

sudo ufw allow 1515/tcp   # Agent registration

sudo ufw allow from 172.31.34.225   # Allow manager

4.3 Enable Logging

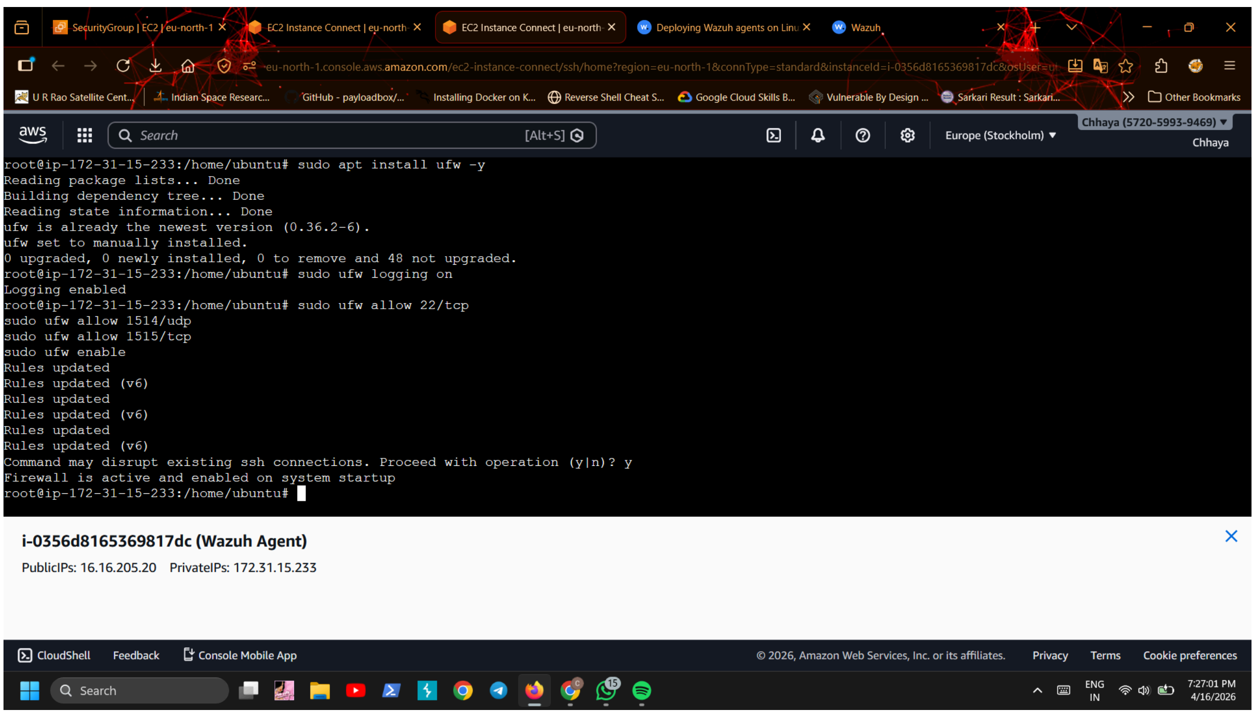
bash

sudo ufw logging on

4.4 Enable Firewall

bash

sudo ufw enable



4.5 Verify

bash

sudo ufw status

5. Log Generation

Firewall logs stored at:

bash

/var/log/ufw.log

Example log: text

[UFW BLOCK] SRC=147.185.133.27 DST=172.31.2.238 DPT=3973

6. Wazuh Agent Configuration

6.1 Configure log monitoring

bash

nano /var/ossec/etc/ossec.conf

Add:

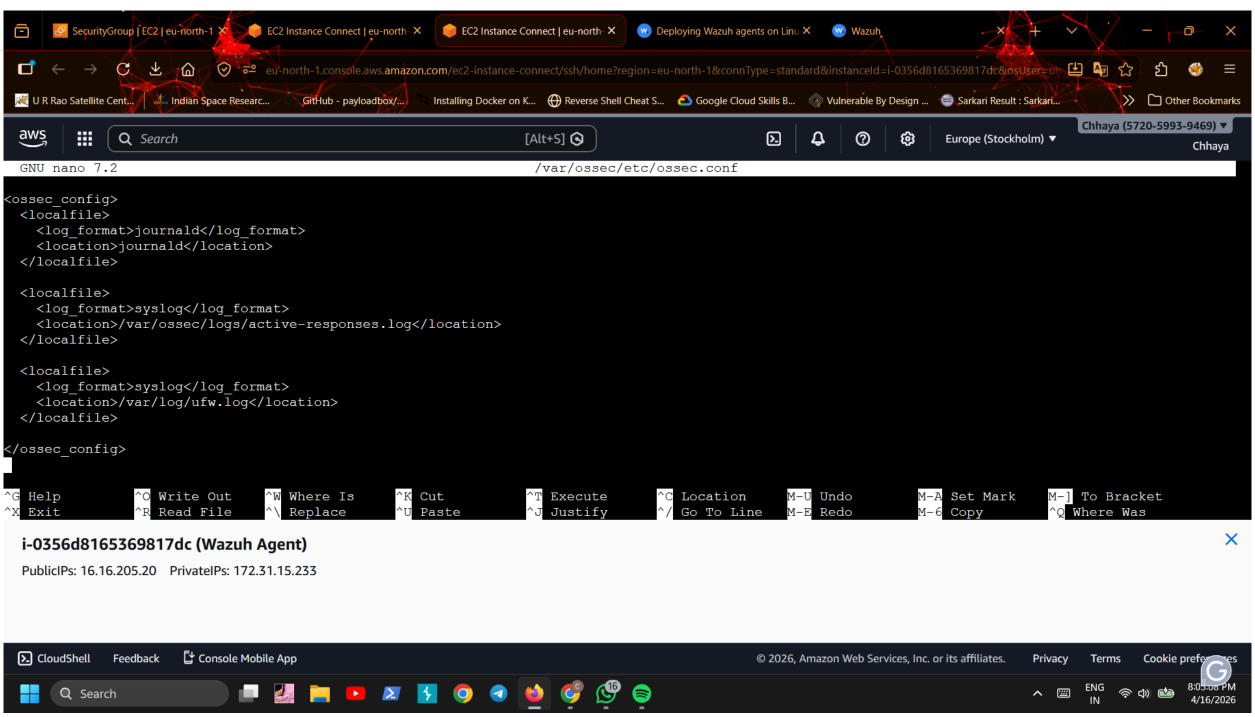
xml

<localfile>

<log_format>syslog</log_format>

<location>/var/log/ufw.log</location>

</localfile>



6.2 Restart agent

bash

```
systemctl restart wazuh-agent
```

7. Log Flow Verification

* Logs appear in Wazuh Dashboard

* Source: agent (172.31.2.238)

* Program: kernel

* Message contains: UFW BLOCK

8. Custom Rule Creation (Detection)

8.1 Create rule

On Wazuh Manager:

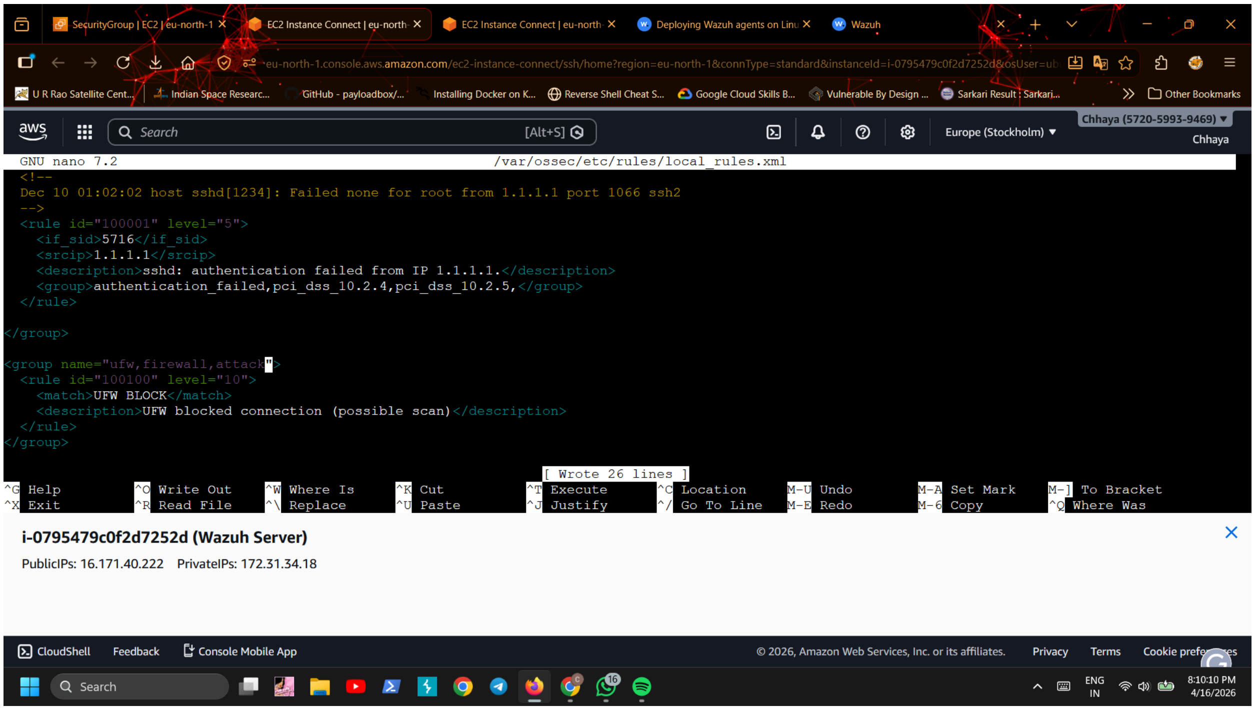
bash

```
nano /var/ossec/etc/rules/local_rules.xml
```

Rule 1: Basic Detection

xml

```
<group name="ufw,firewall,attack">

  <rule id="100100" level="10">

    <match>UFW BLOCK </match>

    <description>UFW blocked connection (possible scan) </description>

  </rule>

</group>
```



8.2 Restart manager

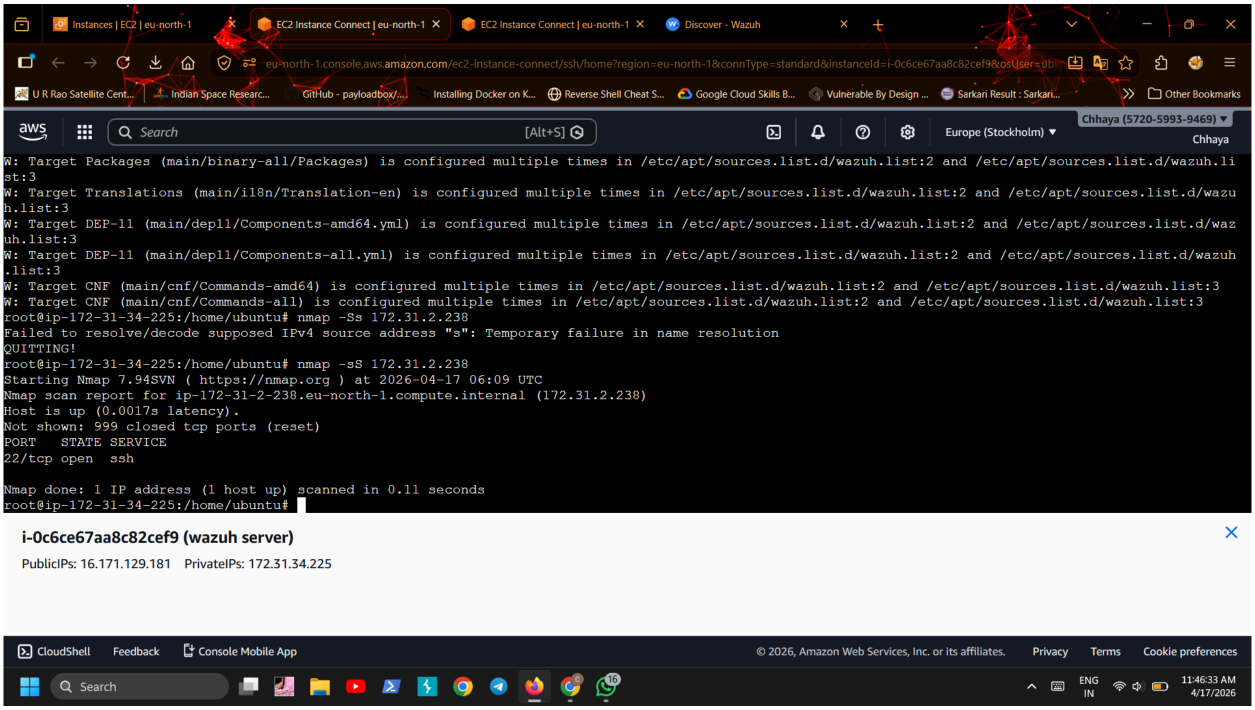
bash

systemctl restart wazuh-manager

9. Testing

From attacker machine:

nmap -sS 172.31.2.238



10. Expected Results

Alert 1:

* Rule ID: 100100

* Description: UFW blocked connection

* Level: 10

11. Troubleshooting Steps

Issue 1: Agent not visible

Fixed by:

bash

agent-auth -m <manager_ip>

Issue 2: Port 1515 blocked

Fixed by:

* AWS Security Group → allow TCP 1515

Issue 3: No UFW logs in Wazuh

Fixed by:

* Adding /var/log/ufw.log in ossec.conf

Issue 4: Rule not triggering

Cause:

* Wrong SID or incorrect rule

Fix:

xml

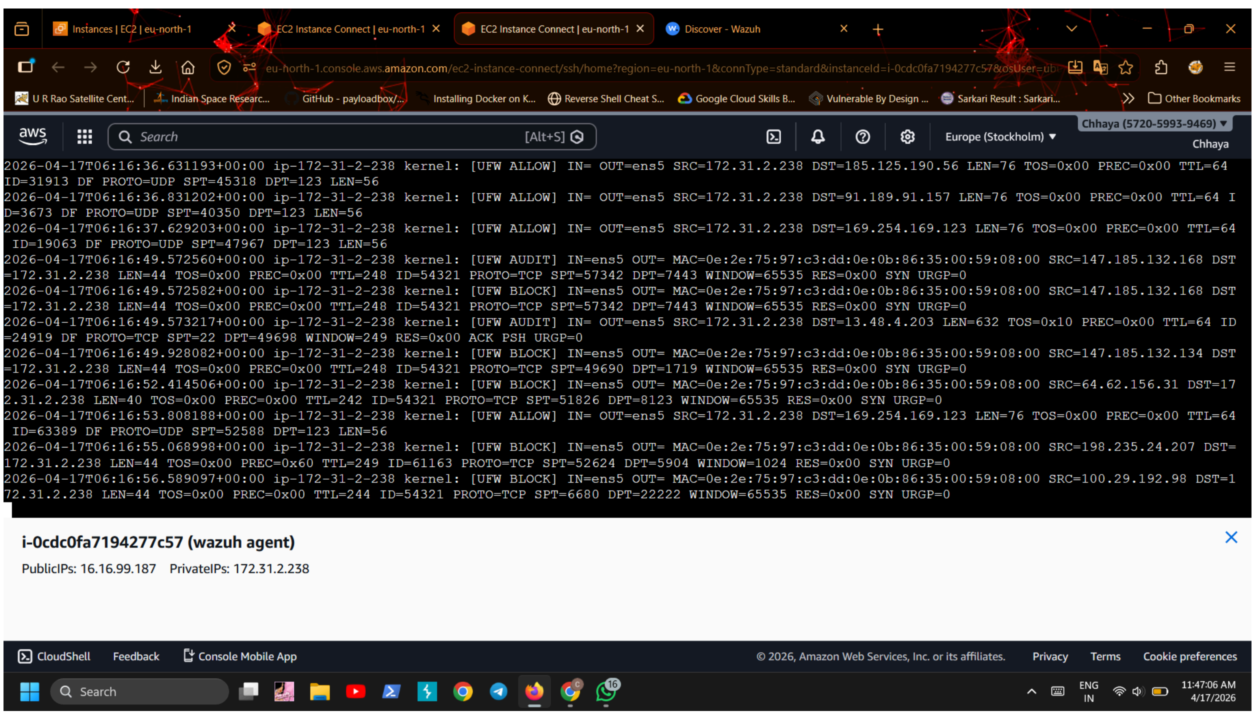
<match>UFW BLOCK</match>

Issue 5: Manager blocked by firewall

Fixed by:

bash

ufw allow from 172.31.2.238



Issue 6: Dashboard “Server not ready”

Fixed by:

* Correct wazuh.yml config

* API authentication

* Clearing cache

12. Key Learnings

* UFW generates firewall logs

* Wazuh parses and correlates logs

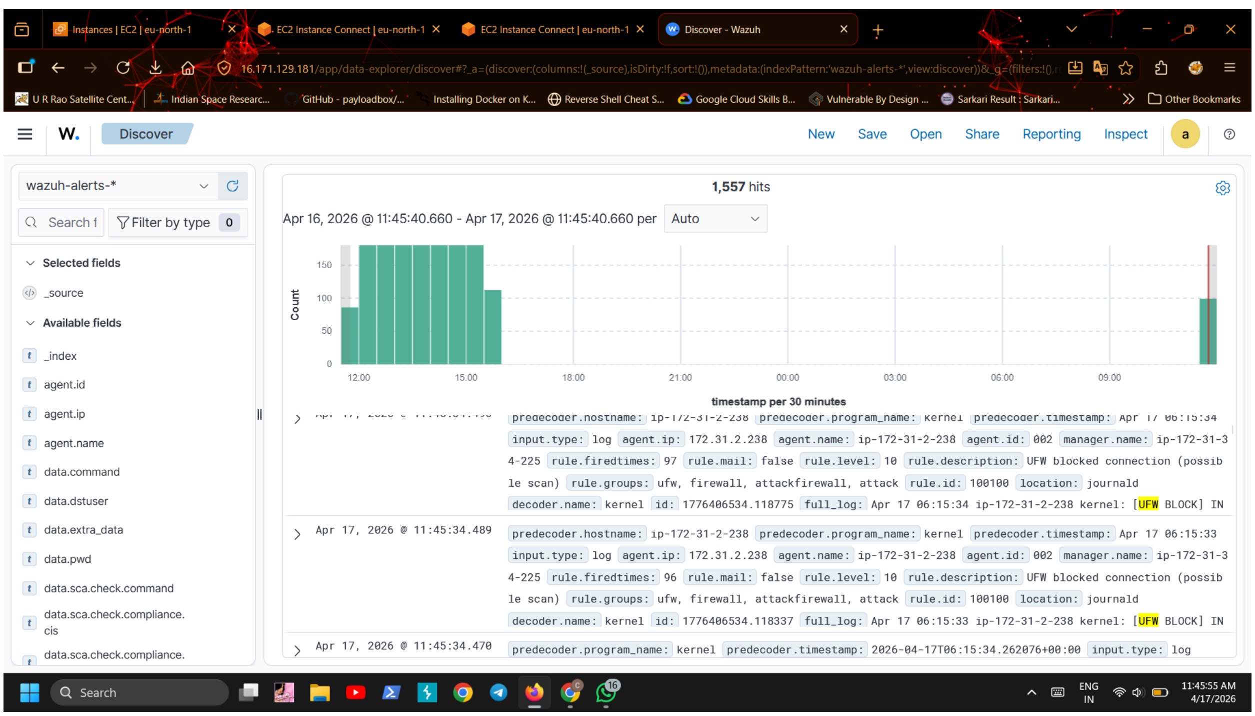
* Custom rules detect threats

* Frequency-based rules detect scans

* Active response enables auto-blocking

13. Final Outcome

- Real-time firewall monitoring
- Detection of external scans
- Alert generation in SIEM
- Capability to auto-block attackers



14. Alerting Configuration (Wazuh Dashboard / OpenSearch)

14.1 Navigate to Alerting

In Wazuh Dashboard:

Go to:

☰ Menu → OpenSearch Plugins → Alerting → Monitors

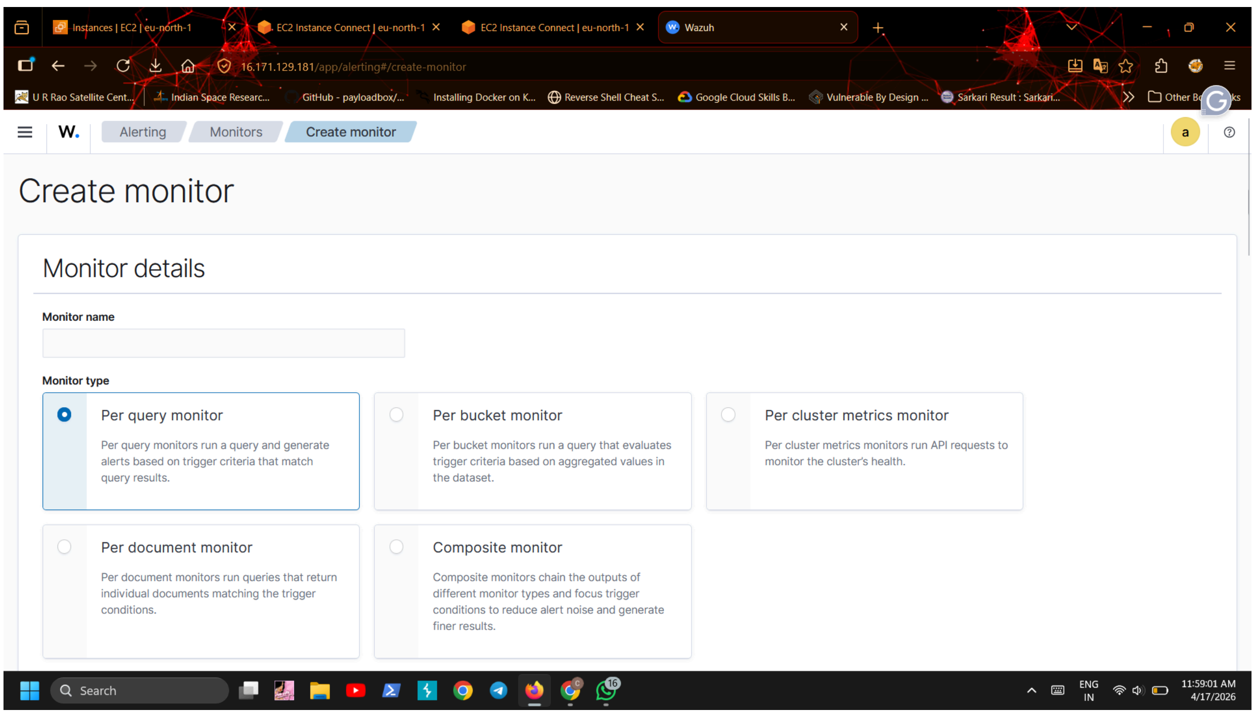
14.2 Create Monitor

Click: *Create monitor*

Select:

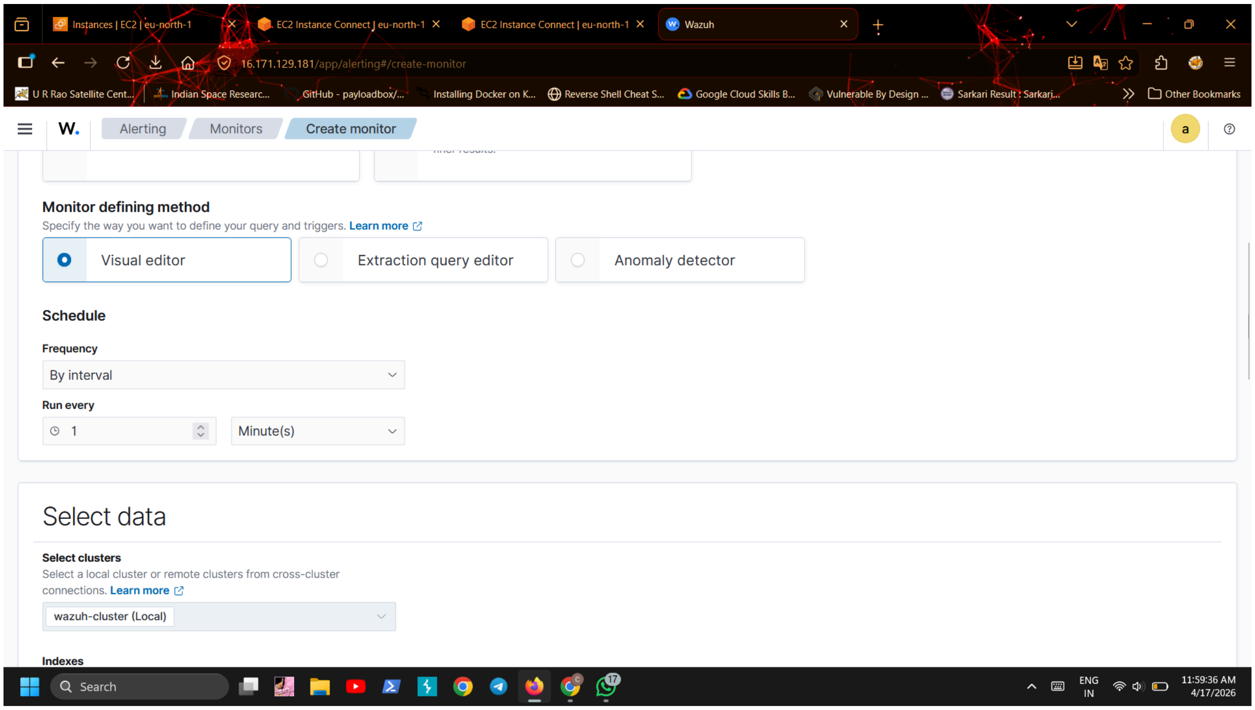
text

Monitor type: Per query monitor



Basic details

Field	Value
Name	UFW Scan Detection Monitor
Index	wazuh-alerts-*
Schedule	Every 1 minute



What this does

* Matches your custom rules:

  * 100100 → single block

  * 100101 → scan detected

14.3 Define Trigger

Click:

Add trigger

Fill:

Field	Value
Trigger name	UFW Attack Detected
Severity	High

Trigger condition

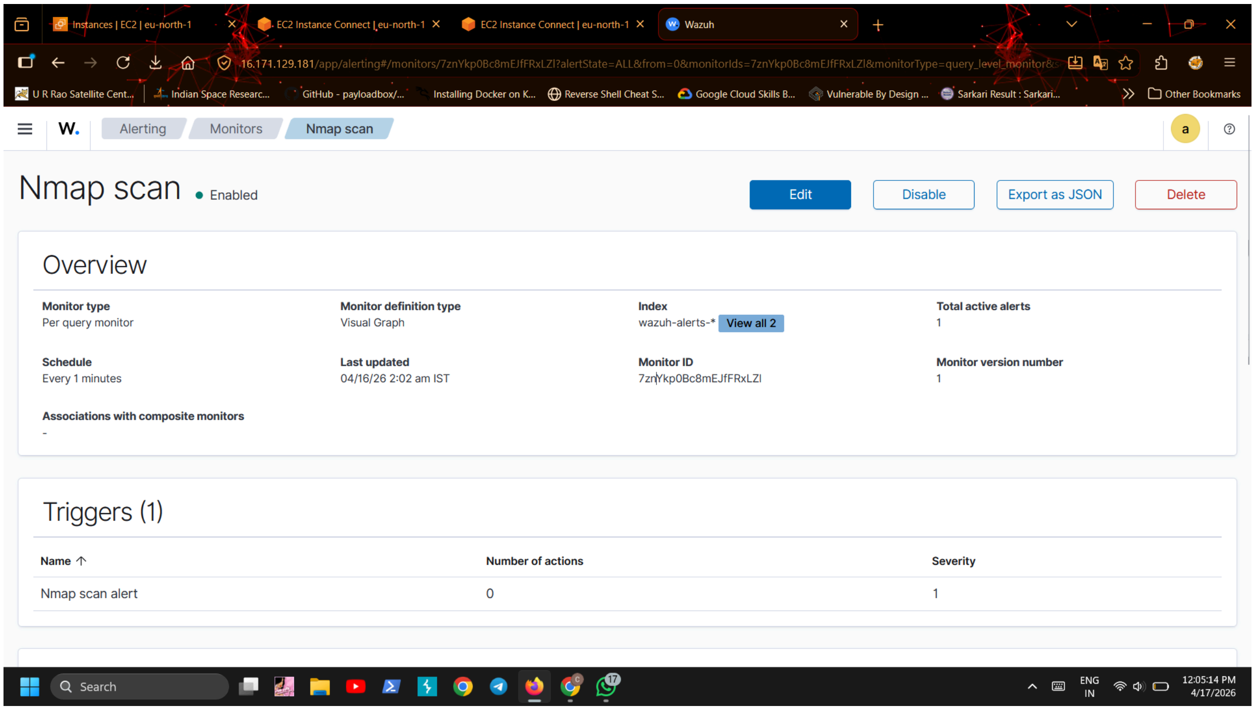
painless

ctx.results[0].hits.total.value > 0

Meaning

If ANY matching alert exists → trigger fires

Create



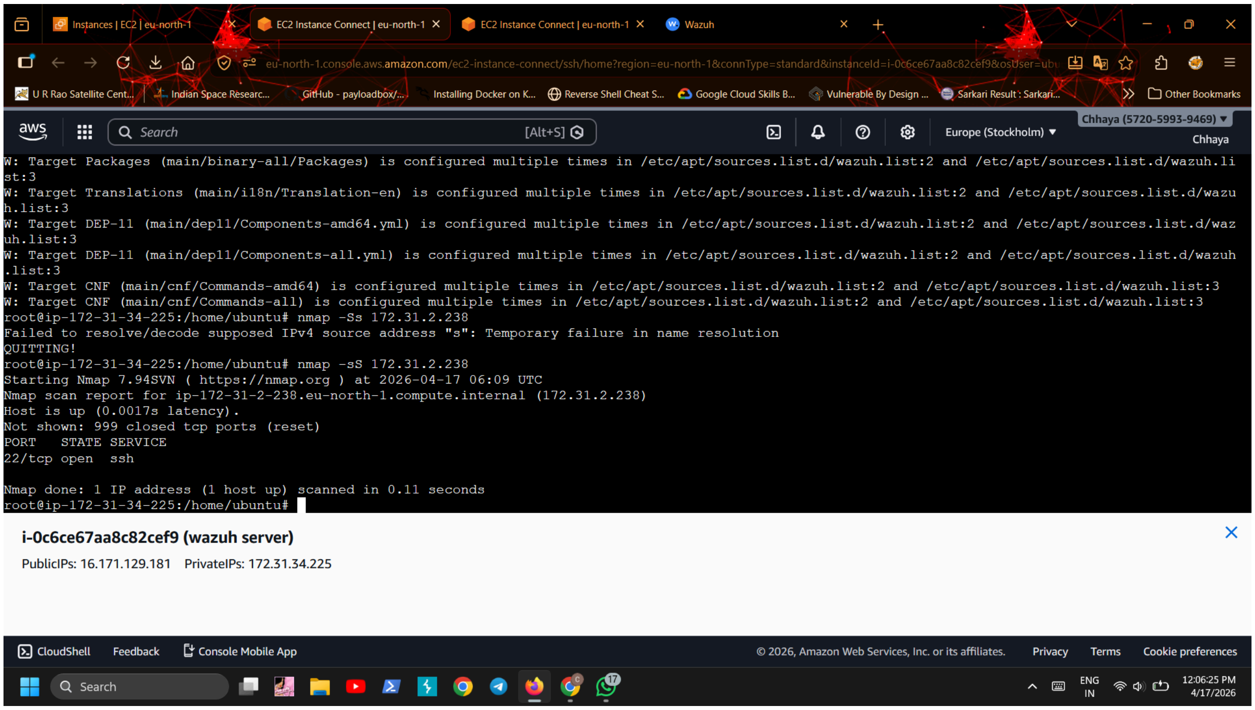
15. Testing Alerting

Step 1 — Trigger attack

From attacker machine:

bash

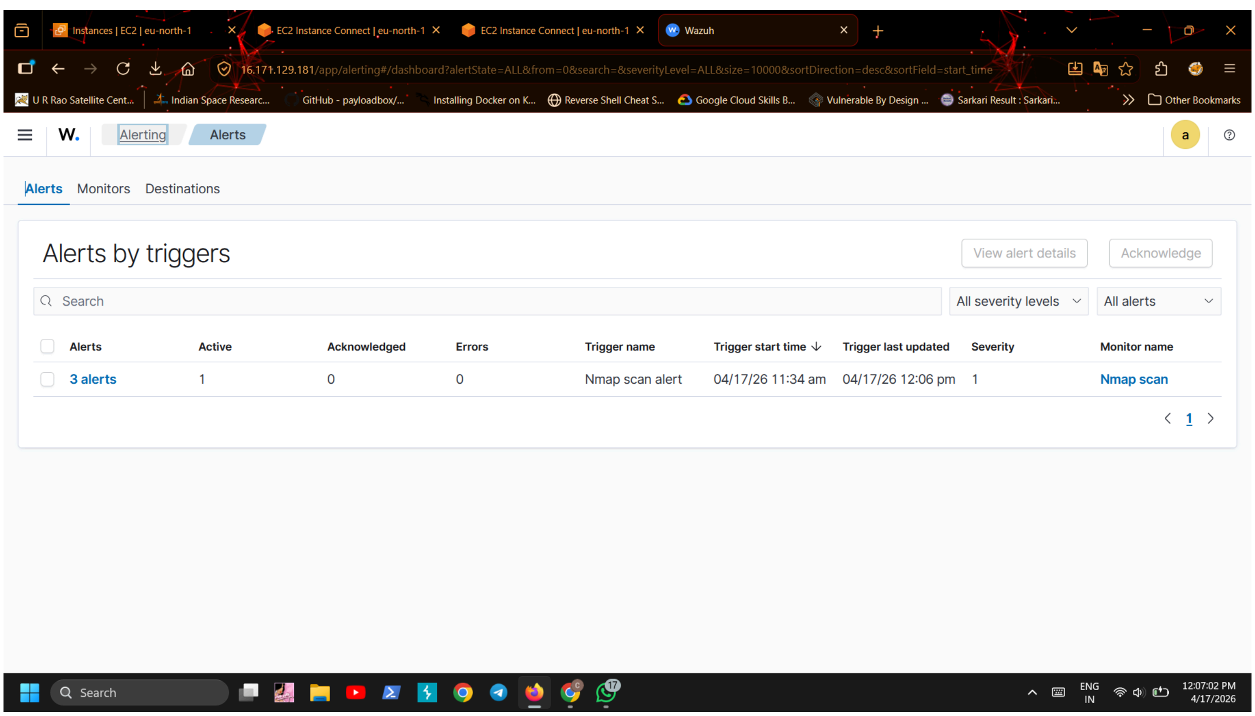
nmap -sS 172.31.2.238



Step 2 — Verify

Go to:

Alerting → Alerts



Expected result

* Monitor triggered

* Alert created

* Severity: High

16. Sample Alert Output

Monitor: UFW Scan Detection Monitor

Trigger: UFW Attack Detected

Rule: Port scan detected (multiple UFW blocks)

Source IP: 147.185.x.x

17. Final Workflow

Attacker →

UFW blocks →

Wazuh detects →

Custom rule fires →

Monitor detects →

Trigger fires →

Alert generated →

Notification sent

Final Summary

A monitoring rule was created in the Wazuh dashboard to detect firewall-based alerts generated from custom rules. When suspicious activity such as port scanning is identified, the system triggers alerts and optionally sends notifications for incident response.